Algebraic Loci of Hodge Classes on Abelian Varieties and K3 Surfaces

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The Hodge conjecture is equivalent to two statements about abelian varieties. We prove this, and we prove that for each statement the set of varieties on which it is known is dense in every family and is either the whole family or a meagre set. We then build the first object each statement needs: a chain of quotients that makes the real multiplication by $\sqrt{6}$ on K3 surfaces algebraic, a complex on a product of two Weil fourfolds with the Chern character the semiregularity criterion asks for, and a single class on an abelian fourteenfold that carries the exceptional classes of a Mumford fourfold. Neither statement is proved. We say at which step each construction stops.

I. INTRODUCTION

The Hodge conjecture asks that every rational cohomology class of type (p,p) on a smooth complex projective variety be a rational combination of classes of algebraic subvarieties.[1] Hodge raised it in 1950 [18]; Lefschetz had proved it for divisors [22]. It is known in a scattered family of cases and open in general. This paper does three things. It proves that the conjecture is equivalent to two statements about abelian varieties. It proves that the locus on which either statement is known has the same shape in every family, dense and either everything or meagre, so that each statement asks that algebraicity propagate from a dense set of special points to the whole family. And it builds, at the first class each statement has to reach, an explicit object, proves what that object does, and names the step at which it stops. The conjecture does not close here. What the paper adds is a complete account of where the two remaining statements live, one new theorem on real multiplication by $\mathbb{Q}(\sqrt{6})$, and three constructions carried to the point where the next step is a single stated question.

A. The two statements

Write **HC** for the Hodge conjecture for all smooth complex projective varieties and **AB** for its restriction to abelian varieties. For a smooth projective variety X and an integer p let $\mathrm{Hdg}^p(X)$ be the Hodge classes of degree $2p$, and let $\mathrm{Ab}^p(X) \subseteq \mathrm{Hdg}^p(X)$ be the span of the classes $\Gamma_*\gamma$, where B runs over the complex abelian varieties (the point included), γ over the Hodge classes on B, and Γ over the algebraic correspondences from B to X. The two statements are

(F2) every Hodge class on every complex abelian variety is algebraic, in the form in which it first appears in the closure computation of [7]: the Hodge classes of the smallest known instance, the exceptional classes on the square of a Mumford fourfold, are algebraic together with everything that can be built from them;

(F3′) $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$ for every X and every p: every Hodge class is algebraic modulo the images, under algebraic correspondences, of Hodge classes of abelian varieties.

Theorem I.1 (Reduction)**.** **HC** *holds if and only if* (F2) *and* (F3′) *both hold. Moreover* (F2) *is equivalent to* **AB**, *and* (F3′) *holds on every abelian variety, on every variety dominated by a product of curves or by an abelian variety, and on every variety in degrees* $2p$ *with* $p \in \{0, 1, \dim X - 1, \dim X\}$.

The content of the theorem is in the second sentence: (F2) is not a special case of the conjecture for abelian varieties but the whole of it, carried by its hardest classes, and (F3′) is exactly what the conjecture asks of the part of the cohomology of X that no abelian variety reaches. Section III proves it and places the two statements in the closure graph of the known reductions, with the Lefschetz standard conjecture, André's motivated classes and the variational Hodge conjecture as the other nodes (Fig. 1).

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[1] An earlier preprint of the authors, *Relative secant cycles and Hodge classes*, Preprints.org (2026), doi:10.20944/preprints202602.0462.v5 [3], claimed a proof of the Hodge conjecture that does not hold, and the present paper replaces it. The secant variety $\mathrm{Sec}^n(A^\vee)$ contains $A^\vee$, so the cycle $\Sigma \cap A^\vee$ of its Definition 3 is $A^\vee$ itself, whose Fourier–Mukai transform is a skyscraper sheaf with no component in degree $2n$, contrary to its Lemma 4. The Weil classes span a 2-dimensional space on which $K^\times$ acts by $\tau \mapsto \tau^{2n}$, while η^n has character $N(\tau)^n$, contrary to its Propositions 1 and 2. Its general case (Theorem 12) applies the Kuga–Satake construction outside Hodge structures of K3 type and assumes the Kuga–Satake correspondence algebraic, which is open. Section VII lists every point at which that preprint fails.

B. The shape of the known locus

For a smooth projective family $\pi\colon \mathcal{X} \to S$ and a flat Hodge class ξ let $\Sigma(\xi) \subseteq S$ be the set of points at which ξ is algebraic.

Theorem I.2 (Dichotomy and propagation)**.**

(i) *$\Sigma(\xi)$ is a countable union of Zariski closed subsets of S, and either $\Sigma(\xi) = S$ or $\Sigma(\xi)$ is meagre.*

(ii) (F2) *holds if and only if, for every polarised abelian scheme and every flat Hodge class ξ, algebraicity of ξ at one point of complex multiplication forces $\Sigma(\xi) = S$.*

(iii) *For the Weil classes of every CM field, every $n \geq 2$ and every discriminant, $\Sigma(\xi)$ contains an explicit point of complex multiplication and is dense.*

(iv) (F3′) *holds for X in degree $2p$ if and only if a Hodge complement of the abelian part of $H^{2p}(X, \mathbb{Q})$ carries no nonzero Hodge class.*

So both statements have the same form. There is a dense set of points, the points of complex multiplication, at which the class is known to be algebraic by explicit cycles; the locus of algebraicity is a countable union of closed strata; and the conjecture is the statement that this union is not meagre. Figure 2 draws it. Section IV proves the theorem, after [7], whose argument we rewrite in the generality needed here.

C. Constructions at the first class

The rest of the paper is constructive. At each statement we go to the first class that is not known and build the object that would make it algebraic.

At (F3′) the first class lives on powers of a K3 surface S with real multiplication: the transcendental lattice $T(S)_{\mathbb{Q}}$ is a vector space over a totally real field $E = \mathrm{End}_{\mathrm{Hdg}} T(S)_{\mathbb{Q}}$, and the elements of E are Hodge classes on $S \times S$. Section V proves:

Theorem I.3 (Chains of symplectic quotients)**.** *Let S be a projective K3 surface and $e \in E$ with $e^2 = n \in \mathbb{Q}_{>0}$. If there is a chain of projective K3 surfaces $S = S_0, S_1, \ldots, S_r$ joined by dominant rational maps or rational Hodge isometries whose degrees multiply to n modulo squares, then e is algebraic on $S \times S$. In particular, for every projective K3 surface S with $\mathbb{Q}(\sqrt{6}) \subseteq E$ whose transcendental space embeds in the invariant lattice of a symplectic automorphism of order three, the real multiplication by $\sqrt{6}$ is algebraic, the Hodge conjecture holds for $S \times S$ and $S^{[2]}$, and such surfaces form two-dimensional families whose very general member has $E = \mathbb{Q}(\sqrt{6})$ exactly.*

Varesco [40] proved the case of a single quotient by a symplectic automorphism of prime order, which gives $\sqrt{2}$ and $\sqrt{3}$; van Geemen and Schütt [44] reach $\sqrt{2}, \sqrt{3}, \sqrt{7}$ through rational self-maps. The chain adds $\sqrt{6}$, the first composite value, with the lattice condition made explicit. A second construction in Section V starts from the universal elliptic curve over the modular curves $X_0(5)$ and $X_0(7)$: the Fricke involution lifts to an isogeny of degree 5 or 7 between the elliptic K3 surface over the base and the one over its image, and on the quadratic twists of the two surfaces by the same pair of parameters it becomes a rational map of that degree between two K3 surfaces of the same family. Theorem V.8 shows that the very general member of each family has Picard number 16 and real multiplication by $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\sqrt{7})$, that the multiplication is the pushforward of the isogeny and so is algebraic, and that the Hodge conjecture holds for the square of every member. The method also shows exactly why it stops. Proposition V.9 describes the action of an arbitrary correspondence on the transcendental lattice: every component whose resolution has geometric genus at most one acts as a similarity, with multiplier the product of the degrees of its two projections, a composite returning to S is an element of E with rational square, and these span only the multiquadratic subfield of E; a correspondence reaching anything else has a component of geometric genus at least two and transcendental rank at least twice that of S. Proposition V.10 shows that the correspondences cut out of finite groups acting on covers of S, which include the graphs of the dihedral families of [44], produce only abelian fields, and Corollary V.11 draws the consequence that the Hodge conjecture holds on $X \times X$ for every member of those families. For the cubic field of the frontier case of [7] the multiquadratic subfield is $\mathbb{Q}$ and the field is not cyclic in general, so both constructions are blind to it.

At (F2) the first class is a Weil class of a very general abelian $2n$-fold of Weil type with $n \geq 3$, or the exceptional class of a Mumford fourfold. For the Weil classes the tool is the polarised semiregularity criterion of [7]: a perfect complex $\mathcal{E}$ whose Chern character is a nonzero multiple of a Weil class plus a polynomial in the polarisation, with $\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$ equal to the lower bound $r(\gamma) = (4+\rho)n^2 - 2n$, makes the Weil classes algebraic on every member with full Hodge group. Section VI proves:

Theorem I.4 (The product object)**.** *Let A, A' be very general polarised abelian varieties of Weil type for the same imaginary quadratic field, of dimensions $2m$, $2m'$, with algebraic Weil classes, and let $n = m + m'$. There is a perfect complex $\mathcal{E}$ on $A \times A'$, a direct sum with shifts of exterior products of secant sheaves and powers of the polarisations, whose Chern character has the polarised shape with Hankel rank $\rho = 3$, so that $r(\gamma) = 7n^2 - 2n$, and whose first-order Hodge locus is the whole tangent space of the polarised Weil family. Every such direct sum has $\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) > r(\gamma)$, and its semiregularity map is not injective. The criterion can therefore be met at a product point only by a convolution of these pieces with nonzero differentials.*

For the exceptional classes of a Mumford fourfold X we replace the K3 surface that carries their transcendental part in [7] by an abelian sixfold Z_K over the same Shimura curve, so that the whole problem stays inside abelian varieties.

Theorem I.5 (The sixfold link)**.** *Let X be a Mumford fourfold with cubic field F, let $K = F(\sqrt{-\delta})$ be a CM extension inside the quaternion algebra, and let Z_K be the abelian sixfold of Shimura type attached to K over the Shimura curve of X. The Hodge classes in $H^1(X)\otimes H^1(X)\otimes H^2(Z_K)$ that identify the two copies of the nine-dimensional representation T form a line over F, spanned by a class ξ. If ξ is algebraic and K contains no imaginary quadratic field, the exceptional classes of $X\times X$ are algebraic, and so is every Hodge class on every power of X. Conversely the algebraicity of ξ is equivalent to the algebraicity of the exceptional classes together with one algebraic link from Z_K to a power of X.*

Neither theorem closes (F2). Section VI says what each needs next: for Theorem I.4 the Ext algebra of Markman's secant sheaves, which no published result computes; for Theorem I.5 the algebraicity of one class in a mixed Künneth component of an abelian fourteenfold, which no known theorem produces even at the points of complex multiplication.

D. What is proved and what is not

Table III in Section VIII lists every statement of the paper with its status. The reduction, the dichotomy, Theorem I.3 and the two constructions are proved. (F2) and (F3′) are open, and with them the conjecture. We state this at the outset because the earlier preprint named in the footnote did not, and because the value of the constructions lies in the precision of the question each one leaves.

Appendix A describes the machine checks: a Lean 4 file that verifies the closure graph, the character computation behind the footnote and the dimension counts of Section VI, with no axiom beyond propositional extensionality; the Lean certificate of [7] on which the reduction rests; and programs in Python, Julia, C, C++ and shell that recompute every finite quantity used in the proofs. No statement of the paper rests on a computation.

II. CONVENTIONS

Varieties are smooth, projective and connected over $\mathbb{C}$ unless said otherwise. Cohomology is singular cohomology with rational coefficients, $H^k(X) = H^k(X,\mathbb{Q})$, with its Hodge structure. A Hodge class of degree $2p$ is an element of $H^{2p}(X)\cap H^{p,p}$; their space is $\mathrm{Hdg}^p(X)$. A class is algebraic if it is a rational combination of fundamental classes of subvarieties of codimension p; by Fulton [14], intersection products, pull-backs and push-forwards along morphisms preserve algebraic classes. An algebraic correspondence from Y to X is an algebraic class Γ on $Y\times X$, acting by $\Gamma_*\gamma = \mathrm{pr}_{X*}([\Gamma]\cup \mathrm{pr}_Y^*\gamma)$. The Hodge group $\mathrm{Hg}(V)$ of a rational Hodge structure V is the smallest algebraic subgroup of $\mathrm{GL}(V)$ defined over $\mathbb{Q}$ whose real points contain the circle $h(U(1))$; for a polarisable V it is reductive, the Hodge classes in every tensor construction on V are its invariants, and the Hodge group of a variety is that of its total cohomology [13]. The Hodge conjecture for X is the statement that every Hodge class on X is algebraic; **HC** is the conjecture for all varieties and **AB** for all abelian varieties.

For an abelian variety A of dimension g with $H^1(A) = V$, the cohomology ring is $\bigwedge^\bullet V$. An abelian $2n$-fold A is of Weil type for the imaginary quadratic field K if K acts on A up to isogeny with V a K-vector space of dimension $2n$ on which the two complex embeddings have equal multiplicity n in $H^{1,0}$. The Weil classes are the elements of $\bigwedge^{2n}_K V \subseteq \bigwedge^{2n} V$, a two-dimensional rational space of Hodge classes of degree $2n$ [45]; $K^\times$ acts on it by $\tau\mapsto\tau^{2n}$, and η^k, for η the class of a K-hermitian polarisation, carries $\tau\mapsto N_{K/\mathbb{Q}}(\tau)^k$. For the general CM field we follow [7]: a family of Weil type for a CM field F of degree $2m$ with $n\geq 2$ and discriminant δ has a connected polarised period domain $\mathcal{D}_F$ of dimension n^2, and its very general member has Hodge group $\mathrm{SU}(V,H)$ for the F-hermitian form H of the polarisation.

A Mumford fourfold is a simple abelian fourfold X whose Hodge group is the $\mathbb{Q}$-form of SL_2^3 given by the norm-one group D^1 of a quaternion algebra D over a totally real cubic field F, split at one real place and definite at the other two [30]. Its Hodge ring is generated by the polarisation and one class of degree four that is not a polynomial in divisors, and $X\times X$ carries exceptional Hodge classes of degree four that are not products of divisor classes and are not Weil classes of any quotient [7, 31]. We write $T = D^0\subseteq D$ for the trace-zero part, a nine-dimensional F-line on which $\mathrm{Hg}(X)$ acts by conjugation; $T(-1)$ is a Hodge structure of K3 type with Hodge numbers $(1,7,1)$, and the exceptional classes of $X\times X$ are the real multiplication by F on the copy of $T(-1)$ inside $H^1(X)\otimes H^1(X)$.

For a K3 surface S the transcendental lattice $T(S)$ is the orthogonal complement of $\mathrm{NS}(S)$ in $H^2(S,\mathbb{Z})$; $T(S)_\mathbb{Q}$ is an irreducible Hodge structure, and $E = \mathrm{End}_{\mathrm{Hdg}}T(S)_\mathbb{Q}$ is a totally real field or a CM field, acting by self-adjoint endomorphisms in the first case [46]. The K3 lattice is $\Lambda = U^3\oplus E_8(-1)^2$, and $\Lambda_\mathbb{Q} = \Lambda\otimes\mathbb{Q}$. For a rational quadratic space W we use the discriminant $d(W)\in\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ and the Hasse invariants $\varepsilon_p(W) = \prod_{i<j}(a_i,a_j)_p$ of a diagonalisation $\langle a_1,\dots,a_k\rangle$, with $(\ ,\ )_p$ the Hilbert symbol; the classification of rational quadratic forms by rank, signature, discriminant and Hasse invariants, and the existence theorem for prescribed local invariants, are Serre's [36, Chap. IV].

Perfect complexes on A are objects of the bounded de-

rived category of coherent sheaves; $\mathrm{ch}(\mathcal{E}) \in H^{\mathrm{ev}}(A)$ is the Chern character, additive on triangles, multiplicative on exterior products, with $\mathrm{ch}(\mathcal{E}[1]) = -\mathrm{ch}(\mathcal{E})$. The semiregularity map $\sigma_{\mathcal{E}} \colon \mathrm{Ext}^2(\mathcal{E},\mathcal{E}) \to \bigoplus_q H^{q+2}(A,\Omega^q)$ is that of Buchweitz and Flenner [5], after Bloch [6]; its injectivity implies that the Hodge locus of $\mathrm{ch}(\mathcal{E})$ coincides, near the given point, with the locus where $\mathcal{E}$ deforms [5, 35].

Throughout, an assertion attributed to a reference is cited and not reproved; everything else is proved in the text. The phrase "very general" means outside a countable union of proper closed analytic subsets of the parameter space.

III. THE REDUCTION TO ABELIAN VARIETIES

A. The statement (F3′) and the class $\mathcal{A}$

Definition III.1. For a variety X and $p \geq 0$ let $\mathrm{Ab}^p(X) \subseteq H^{2p}(X)$ be the span of the classes $\Gamma_*\gamma$ of degree $2p$, where B runs over the complex abelian varieties, the point included, γ over $\mathrm{Hdg}^*(B)$, and Γ over the algebraic cycles on $B \times X$ with rational coefficients. The statement (F3′) is: $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$ for every X and every p.

Each $\Gamma_*\gamma$ is a Hodge class, since cup product with an algebraic class and push-forward along a projection are morphisms of Hodge structures of the right bidegree; so $\mathrm{Ab}^p(X) \subseteq \mathrm{Hdg}^p(X)$. Taking B a point shows that $\mathrm{Ab}^p(X)$ contains every algebraic class, and taking X abelian, $B = X$ and Γ the diagonal shows that (F3′) holds on every abelian variety.

Proposition III.2. **HC** *holds if and only if* **AB** *and* (F3′) *both hold.*

Proof. If **HC** holds then **AB** holds, and every Hodge class is algebraic, hence in $\mathrm{Ab}^p(X)$. Conversely let $\gamma \in \mathrm{Hdg}^p(X)$. By (F3′), $\gamma = \sum_j \Gamma_{j*}\gamma_j$ with γ_j a Hodge class on an abelian variety B_j, algebraic by **AB**; then $\mathrm{pr}_{B_j}^*\gamma_j \cup [\Gamma_j]$ is algebraic on $B_j \times X$ and so is its push-forward to X. □

Proposition III.3. *Let $\mathcal{A}$ be the class of varieties X such that $H^*(X)$ is spanned by classes $\Gamma_* y$ with B abelian, Γ an algebraic cycle on $B \times X$ and $y \in H^*(B)$.*

(i) *Every $X \in \mathcal{A}$ satisfies* (F3′).

(ii) *$\mathcal{A}$ contains the abelian varieties and the curves, and is closed under products and under images of surjective morphisms. It contains every product of curves, every symmetric product of a curve, and every variety dominated by a product of curves or by an abelian variety.*

(iii) *Every X of dimension d satisfies* (F3′) *in the degrees $2p$ with $p \in \{0, 1, d-1, d\}$. In particular every variety of dimension at most three satisfies it in every degree.*

Proof. (i) Let $\gamma \in \mathrm{Hdg}^p(X)$ and write $\gamma = \sum_j \Gamma_{j*} y_j$ with $y_j \in H^*(B_j)$. Each Γ_{j*} is a morphism of Hodge structures, so after replacing y_j by its component in the degree that Γ_{j*} sends to $2p$ we may assume y_j homogeneous. Let π_j be the projector of $H^*(B_j)$ onto its Hodge classes along a complement stable under the Hodge group, which exists because that group is reductive. Then $\gamma = \sum_j \Gamma_{j*}\pi_j y_j + \sum_j \Gamma_{j*}(1-\pi_j)y_j$. The second sum is a Hodge class whose summands lie in images of Hodge complements of the invariants; the Hodge group of $\prod_j B_j \times X$ acts on each term through a representation without invariants, so each term, being a Hodge class, is zero. Hence $\gamma = \sum_j \Gamma_{j*}(\pi_j y_j) \in \mathrm{Ab}^p(X)$.

(ii) The point, abelian varieties and curves are in $\mathcal{A}$: for a curve C, H^0 and H^2 come from the point and $H^1(C) = H^1(J(C))$ by the Abel–Jacobi map. If $X, Y \in \mathcal{A}$ then $H^*(X \times Y) = H^*(X) \otimes H^*(Y)$ is spanned by exterior products of classes $\Gamma_* y \boxtimes \Gamma'_* y' = (\Gamma \times \Gamma')_*(y \boxtimes y')$ with $B \times B'$ abelian. If $f \colon X \to X'$ is surjective, then $f_* f^* = \deg(f)$ on $H^*(X')$ after cutting X by hyperplanes to the dimension of X' if needed, so every class of X' is f_* of a class of X, and $f_*\Gamma_* y = (\Gamma_f \circ \Gamma)_* y$ with Γ_f the graph of f. The listed varieties follow: a symmetric product is the image of a product of curves, and a variety dominated by a product of curves or an abelian variety is such an image.

(iii) Degrees 0 and $2d$ are spanned by the class of X and the class of a point. In degree 2 every Hodge class is a divisor class by Lefschetz [22]. In degree $2d-2$, let θ be a hyperplane class; the hard Lefschetz theorem makes $\theta^{d-2}\cup \colon H^2(X) \to H^{2d-2}(X)$ an isomorphism of Hodge structures, so every Hodge class of degree $2d-2$ is $\theta^{d-2} \cup \alpha$ with α a Hodge class of degree 2, algebraic by Lefschetz; a product of algebraic classes is algebraic. □

B. The statement (F2)

For an abelian variety Y let $R(Y) \subseteq H^*(Y)$ be the $\mathbb{Q}$-subalgebra generated by the divisor classes of Y and by the pull-backs f^*w of Weil classes w of abelian varieties B of Weil type along homomorphisms $f \colon Y \to B$. Every element of $R(Y)$ is a Hodge class, and every Hodge class on an abelian variety of dimension at most five lies in $R(Y)$ [32].

Definition III.4. The statement (F2) is: for every complex abelian variety Y, every Hodge class on Y that does not lie in $R(Y)$ is algebraic.

The classes in $R(Y)$ are the ones the known theorems reach when they reach anything: divisor classes by Lefschetz, Weil classes by Markman [25–27] on fourfolds of Weil type and by the propagation results of [7] wherever they propagate. So (F2) is the conjecture for abelian varieties with the known classes struck out. The next proposition says that striking them out removes nothing.

Setting III.5. Let X be a Mumford fourfold, $V = H^1(X)$ with its polarisation ψ, $G = \mathrm{Hg}(X) \subseteq \mathrm{Sp}(V,\psi)$, and $e_0 \in \mathrm{Hdg}^2(X\times X)$ an exceptional class, a Hodge class of degree four that is not a polynomial in divisor classes. Such classes exist [31]; X is simple with $\mathrm{End}^0(X) = \mathbb{Q}$ [30].

Lemma III.6. *Let Y be an abelian variety. Let $Y_X \subseteq Y$ be the largest abelian subvariety isogenous to a power of X and Y' the identity component of the intersection of the kernels of the homomorphisms $Y \to X$. Then $Y_X \times Y' \to Y$ is an isogeny,* $\mathrm{Hom}(X,Y') = 0$, *and*

$$H^1(Y) = \big(V \otimes Q\big) \oplus H^1(Y'), \qquad Q = \mathrm{Hom}^0(Y_X, X),$$

the first summand embedded by $v\otimes f \mapsto f^*v$. Let $\mathrm{Sp}(V,\psi)$ act on $H^*(Y) = \bigwedge^* H^1(Y)$ through the factor V. Then every class of $R(Y)$ is invariant under $\mathrm{Sp}(V,\psi)$, and for $Y = X^m$ the invariants of $\mathrm{Sp}(V,\psi)$ in $H^*(X^m)$ are polynomials in divisor classes.

Proof. Poincaré reducibility gives $Y \sim Y_X \times Y'$ with $Y_X \sim X^k$ and no factor X in Y'; since X is simple, $\mathrm{Hom}(X,Y') = \mathrm{Hom}(Y',X) = 0$, and since $\mathrm{End}^0(X) = \mathbb{Q}$, Q has dimension k and the displayed map is an isomorphism. A homomorphism $h\colon Y \to Z$ of abelian varieties decomposes up to isogeny as $h_X \times h'$, and h^* acts on H^1 as $(1 \otimes (f \mapsto f \circ h_X)) \oplus h'^*$, which commutes with the action of $\mathrm{GL}(V)$ on the factor V. So pull-backs along homomorphisms, cup products and exterior products are $\mathrm{GL}(V)$-equivariant, and the classes invariant under $\mathrm{Sp}(V,\psi)$ form a subalgebra stable under pull-back. It remains to check the generators of $R(Y)$.

A divisor class on Y has Künneth components in $\bigwedge^2(V\otimes Q)$, $(V\otimes Q)\otimes H^1(Y')$ and $\bigwedge^2 H^1(Y')$, each a Hodge class. The middle one is a homomorphism from Y' to the dual of Y_X, which is zero. The first is a Hodge class on Y_X, hence a G-invariant of $\bigwedge^2(V\otimes Q) = S^2V \otimes \bigwedge^2 Q \oplus \bigwedge^2 V \otimes S^2Q$. Since G contains the Hodge group of a simple abelian fourfold whose only Hodge classes of degree two are the divisor classes, $(S^2V)^G = 0$ and $(\bigwedge^2 V)^G = \mathbb{Q}\psi$; so the component lies in $\psi \otimes S^2Q$, which $\mathrm{Sp}(V,\psi)$ fixes. The third component is fixed because $\mathrm{GL}(V)$ acts trivially on $H^1(Y')$.

For a Weil class, let B be of Weil type for K and $f\colon Y \to B$. The field K acts on $B \sim B_X \times B'$ through $\mathrm{End}^0(B_X) \times \mathrm{End}^0(B')$, because $\mathrm{Hom}(B_X,B') = 0$; so it acts on $Q_B = \mathrm{Hom}^0(B_X,X)$ and on $H^1(B')$, and the eigenspace of $H^1(B,\bar{\mathbb{Q}})$ for an embedding σ of K is $V\otimes Q_{B,\sigma} \oplus H^1(B',\bar{\mathbb{Q}})_\sigma$. A Weil class lies in the sum over σ of the top exterior powers of these eigenspaces, and $\bigwedge^{\mathrm{top}}(V\otimes Q_{B,\sigma} \oplus W_\sigma) = (\det V)^{\otimes \dim Q_{B,\sigma}} \otimes (\det Q_{B,\sigma})^{\otimes 8} \otimes \det W_\sigma$, on which $\mathrm{SL}(V) \supseteq \mathrm{Sp}(V,\psi)$ acts trivially. Pulling back along f preserves invariance.

For $Y = X^m$, $H^*(X^m) = \bigwedge^*(V\otimes\mathbb{Q}^m)$, and the invariants of $\mathrm{Sp}(V,\psi)$ in the tensor algebra of V are spanned by complete contractions with ψ by the first fundamental theorem for the symplectic group [17, Chap. 5]; a contraction of two factors of X^m is the pull-back of the class $\psi \in H^2(X\times X)$ along the projection to those factors, a divisor class, and a contraction within one factor is $\psi \in H^2(X)$. So the invariants are polynomials in divisor classes. □

Theorem III.7. (F2) *implies* **AB**, *and is therefore equivalent to it.*

Proof. **AB** contains (F2). Assume (F2). By Lemma III.6 the exceptional class e_0 is not invariant under $\mathrm{Sp}(V,\psi)$, since it is not a polynomial in divisor classes; as $\mathrm{Sp}(V,\psi)(\mathbb{Q})$ is Zariski dense in $\mathrm{Sp}(V,\psi)$, there is $g \in \mathrm{Sp}(V,\psi)(\mathbb{Q})$ with $ge_0 \neq e_0$. Let θ be an ample class on $X\times X$, fixed by g by the lemma.

Let A be an abelian variety and $\beta \neq 0$ a Hodge class of degree $2p$ on A, and put $Y = A \times X \times X$. The group $\mathrm{Sp}(V,\psi)$ acts on $H^*(Y) = H^*(A)\otimes H^*(X\times X)$ diagonally, through the copies of V in both factors; on $H^*(A)$ it acts through A_X and fixes β when $\beta \in R(A)$, but we need only that $g(\beta\otimes e_0) = g\beta \otimes ge_0$. If $\beta\otimes e_0$ lay in $R(Y)$ it would be fixed by g, so $g\beta\otimes ge_0 = \beta\otimes e_0$, which forces $ge_0 \in \mathbb{Q}e_0$; and g preserves the polarisation of $X\times X$, hence the intersection form on $H^4(X\times X)$ in which e_0 has nonzero square (the Hodge–Riemann relations on the primitive Hodge classes, after subtracting the part of e_0 in $\theta \cup H^2$, which is a polynomial in divisor classes and may be dropped from e_0 at the start), so $ge_0 = \pm e_0$, and replacing g by g^2 if needed, $ge_0 = e_0$, a contradiction. So $\beta\otimes e_0 \notin R(Y)$ and (F2) makes it algebraic. In the same way $e_0 \cup \theta^4 \notin R(X\times X)$: otherwise $(ge_0 - e_0)\cup\theta^4 = 0$, and $\theta^4\cup\colon H^4(X\times X) \to H^{12}(X\times X)$ is injective by hard Lefschetz. So (F2) makes every Hodge class of degree twelve on $X\times X$ algebraic.

The Hodge group of $X\times X$ is reductive, the Hodge classes of degrees four and twelve are its invariants, and Poincaré duality is an equivariant perfect pairing between H^4 and H^{12}; so it restricts to a perfect pairing between Hdg^2 and Hdg^6, and there is a Hodge class e' of degree twelve with $\int_{X\times X} e_0 \cup e' = 1$. Then

$$\beta = \mathrm{pr}_{A*}\big((\beta\otimes e_0) \cup \mathrm{pr}^*_{X\times X} e'\big),$$

a push-forward of a product of algebraic classes, is algebraic. □

Theorem I.1 is Propositions III.2, III.3 and Theorem III.7 together.

C. The closure graph

The paper [7] records the implications between the conjecture and the statements that have been used to approach it: the Lefschetz standard conjecture (L) for all varieties, the conjecture (M) that every Hodge class is motivated in André's sense [2], the variational Hodge conjecture (V), the statement (F3) that the conjecture holds for all varieties that are not abelian, and the two statements of this paper. The rules are listed in Table I,

TABLE I. The rule set. A rule with two premises needs both. The closure of a set of statements under these rules, and the list of minimal sets whose closure contains **HC**, are computed by hand in the text and checked in Lean (Appendix A).

premises	conclusion	proof
HC	(F3), (F2), (F3′), **AB**, (M)	definitions; [2]
(F3)	**HC**	[7, Prop. 20.33]
(F2)	**AB**	Theorem III.7
AB	(F2)	definition
AB, (F3′)	**HC**	Proposition III.2
(L), (M)	**HC**	[2]; [7, Thm. 20.70]
(L)	**AB**	[7, Thm. 20.70]
(V)	**AB**	[7, Thm. 20.70]

each with the place where it is proved; the first two rows are Propositions III.2 and Theorem III.7.

Proposition III.8. *Under the rules of Table I the closure of a set of statements contains* **HC** *if and only if the set contains one of*

$$\{\mathrm{F3}\},\ \{\mathrm{L},\mathrm{M}\},\ \{\mathrm{L},\mathrm{F3}'\},\ \{\mathrm{V},\mathrm{F3}'\},\ \{\mathrm{F2},\mathrm{F3}'\},\ \{\mathbf{AB},\mathrm{F3}'\},$$

and these six sets are the minimal ones with this property.

Proof. Each listed set derives **HC**: {F3} by the second row; {L, M} by the sixth; the other four through **AB**, which (F2), (L) and (V) each give, and the fifth row. For the converse, call a set closed if it is stable under the rules. A closed set without **HC** contains no (F3), does not contain (L) and (M) together, and, if it contains (F3′), contains none of **AB**, (F2), (L), (V), since each of these yields **AB** and then **HC**. The maximal closed sets without **HC** are therefore

$$M_1 = \{\mathrm{F2}, \mathbf{AB}, \mathrm{L}, \mathrm{V}\},$$
$$M_2 = \{\mathrm{F2}, \mathbf{AB}, \mathrm{V}, \mathrm{M}\},$$
$$M_3 = \{\mathrm{F3}', \mathrm{M}\},$$

each of which is closed by inspection of the table. The closure of a set Σ contains **HC** exactly when Σ lies in none of the M_i. If Σ contains (F3) this holds. Otherwise Σ needs an element outside M_1, that is (F3′) or (M); one outside M_2, that is (F3′) or (L); and one outside M_3, that is one of (F2), **AB**, (L), (V). The minimal choices are (F3′) with one of (F2), **AB**, (L), (V), and (M) with (L). This gives the six sets, and each is minimal because removing any element lands in some M_i. The same computation is carried out mechanically from the rule list in the Lean file of Appendix A, which returns these six sets and no others. □

Figure 1 draws the graph. The two statements of this paper are the nodes marked in colour; every route to **HC** other than (F3), which is **HC** itself, and (L)+(M) passes through (F3′).

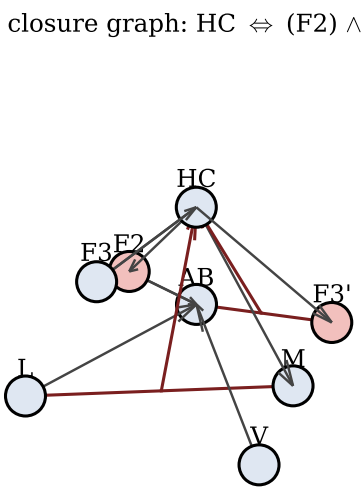


FIG. 1. The closure graph of Table I. Grey arrows are rules with one premise; a red fork joins the two premises of a rule that needs both. The coloured nodes are (F2) and (F3′).

IV. THE ALGEBRAIC LOCUS IN A FAMILY

Definition IV.1. Let $\pi\colon \mathcal{X} \to S$ be a smooth projective morphism to a smooth connected complex quasi-projective variety, and let ξ be a global section of $R^{2p}\pi_*\mathbb{Q}$ that is a Hodge class at one point. The algebraic locus of ξ is the set $\Sigma(\xi)$ of $s \in S$ at which $\xi_s \in H^{2p}(\mathcal{X}_s)$ is algebraic.

By the theorem of the fixed part [1, 11], the flat sections of a polarisable variation form a sub-Hodge structure, constant along S; so ξ_s is a Hodge class at every point once it is one somewhere.

Lemma IV.2 (Dichotomy)**.**

(i) $\Sigma(\xi)$ *is a countable union of Zariski closed subsets of* S.

(ii) *Either* $\Sigma(\xi) = S$ *or* $\Sigma(\xi)$ *is meagre in* S^{an}.

(iii) *The following are equivalent:* $\Sigma(\xi) = S$; $\Sigma(\xi)$ *has an interior point;* $\Sigma(\xi)$ *is not meagre;* ξ_s *is algebraic for the very general* s. *If* $\Sigma(\xi)$ *is dense they are equivalent to* $\Sigma(\xi)$ *being closed.*

Proof. (i) Fix a relatively ample line bundle on $\mathcal{X}$. For a Hilbert polynomial P the relative Hilbert scheme $\mathrm{Hilb}^P(\mathcal{X}/S)$ is projective over S [15], and the class of the fibre of the universal family in $H^{2p}(\mathcal{X}_s)$ is locally constant on each connected component of it, since a flat family of subschemes over a connected base has locally constant cycle class. A rational combination $\sum_i a_i[Z_i]$ equal to ξ_s is a point of a fibre product $\prod_S \mathrm{Hilb}^{P_i}(\mathcal{X}/S)$, projective

over S and with countably many connected components; on each component the condition $\sum a_i[Z_i] = \xi$ is either empty or holds throughout, because both sides are flat sections. So $\Sigma(\xi)$ is the union, over the countably many tuples (P_i, a_i) and components, of images of proper morphisms to S, each Zariski closed.

(ii) A countable union of proper Zariski closed subsets of an irreducible S has empty interior in S^{an}, and each is closed, so the union is meagre; if one of the closed sets is S itself, $\Sigma(\xi) = S$.

(iii) By the Baire category theorem S^{an} is not meagre in itself, so $\Sigma(\xi) = S$ excludes meagreness, and a set with an interior point is not meagre; by (ii) not meagre forces $\Sigma(\xi) = S$. The complement of a countable union of proper closed subvarieties is the very general locus, so algebraicity at the very general point means that one of the closed sets is all of S. A dense set that is also closed is S. □

Lemma IV.3 (Points of CM type are dense)**.** *Let G be a connected reductive group over $\mathbb{Q}$, $u_0\colon U(1) \to G_{\mathbb{R}}$ a homomorphism and $\mathcal{D}$ the orbit of u_0 under conjugation by $G(\mathbb{R})^+$. The points $u \in \mathcal{D}$ for which $u(U(1))$ lies in the real points of a maximal torus of G defined over $\mathbb{Q}$ are dense in $\mathcal{D}$.*

Proof. Let $u_1 \in \mathcal{D}$ and W a neighbourhood of u_1. A maximal torus T_1 of $G_{\mathbb{R}}$ contains $u_1(U(1))$, since the centraliser of that compact torus is reductive and connected. Pick a regular element Y_1 in the Lie algebra $\mathfrak{t}_1$ of T_1. The map $(g, Y) \mapsto \mathrm{Ad}(g)Y$ from $G(\mathbb{R})^+ \times \mathfrak{t}_1$ to $\mathfrak{g}_{\mathbb{R}}$ is submersive at $(1, Y_1)$ because $\mathfrak{g}_{\mathbb{R}} = \mathfrak{t}_1 \oplus [\mathfrak{g}_{\mathbb{R}}, Y_1]$, so the image of a small neighbourhood $N_1 \times N_2$, with $N_1 u_1 N_1^{-1} \subseteq W$ and N_2 made of regular elements, contains an open set of $\mathfrak{g}_{\mathbb{R}}$, hence a rational point $Y = \mathrm{Ad}(g)Y'$. The identity component T of the centraliser of the rational regular semisimple Y is a maximal torus over $\mathbb{Q}$ with $T_{\mathbb{R}} = gT_1g^{-1}$, and $u = gu_1g^{-1} \in W$ has $u(U(1)) \subseteq T_{\mathbb{R}}$. □

When $G = \mathrm{Hg}(A)$ and u is the Hodge cocharacter of an abelian variety A, the condition of the lemma says that $\mathrm{Hg}(A)$ lies in a $\mathbb{Q}$-torus, and then A is of CM type: a maximal $\mathbb{Q}$-torus of $\mathrm{GL}(H^1(A))$ containing it is the unit group of a commutative semisimple algebra L of dimension $2\dim A$ commuting with $\mathrm{Hg}(A)$, and the commutant of $\mathrm{Hg}(A)$ is $\mathrm{End}^0(A)$.

Proposition IV.4 (A family with a CM member through every Hodge class)**.** *Let A_0 be an abelian variety and γ a Hodge class on it. There are a polarised abelian scheme $\pi\colon \mathcal{A} \to S$ over a smooth connected quasi-projective base, a flat section ξ of $R^{2p}\pi_*\mathbb{Q}$, a point s_0 with $\mathcal{A}_{s_0} \cong A_0$ and $\xi_{s_0} = \gamma$, and a point c with $\mathcal{A}_c$ of CM type; the points of CM type are dense in S.*

Proof. Let $G = \mathrm{Hg}(A_0)$, let u_0 be the Hodge cocharacter of A_0 and $\mathcal{D}$ its $G(\mathbb{R})^+$-orbit. Every $u \in \mathcal{D}$ defines a polarisable Hodge structure of weight one on $H^1(A_0)$ with Hodge group contained in G, so an abelian variety A_u up to isogeny; fixing the lattice $H^1(A_0, \mathbb{Z})$, the polarisation of A_0 and a level structure of level $N \geq 3$, the quotient $S = \Gamma\backslash\mathcal{D}$ by a neat congruence subgroup $\Gamma \subseteq G(\mathbb{Q})$ is a smooth quasi-projective variety carrying a polarised abelian scheme with these fibres [11, 30]. The class γ is invariant under G, so it is a flat section ξ of $R^{2p}\pi_*\mathbb{Q}$, and $\xi_{s_0} = \gamma$ at the image s_0 of u_0. By Lemma IV.3 the points of CM type are dense in $\mathcal{D}$, hence in S. □

Theorem IV.5 ((F2) is propagation from CM points)**.** *The following are equivalent.*

(i) (F2).

(ii) **AB**.

(iii) *For every polarised abelian scheme $\pi\colon \mathcal{A} \to S$ and every flat Hodge class ξ, algebraicity of ξ_s at one point s forces $\Sigma(\xi) = S$.*

(iv) *The same for the points s of CM type.*

(v) *For every π, ξ with ξ_c algebraic at a point c of CM type, $\Sigma(\xi)$ has an interior point.*

Proof. (i) and (ii) are Theorem III.7. If (ii) holds, ξ_t is a Hodge class at every t by the theorem of the fixed part and algebraic by (ii), so (iii) holds; (iv) is a case of (iii), and (iv) and (v) are equivalent by Lemma IV.2(iii).

Assume (iv). Let F be a CM field and $n \geq 2$. The family of abelian varieties of split Weil type for F over its Shimura variety is a polarised abelian scheme on which the Weil classes are flat sections, and by [7, Thm. 1.1(i)] they are algebraic at a member isogenous to a product of abelian varieties with complex multiplication by F, a member of CM type; so by (iv) they are algebraic on every member, and by pull-back along an F-linear isogeny on every abelian variety of split Weil type for F. By André's theorem, in the form Milne gives it [28, Thm. 1 and §5], every Hodge class on an abelian variety of CM type is a sum of pull-backs, along homomorphisms, of Weil classes of abelian varieties of split Weil type; so **AB** holds for every abelian variety of CM type. Now let A_0 be any abelian variety and γ a Hodge class on it, and take π, ξ, s_0, c from Proposition IV.4. Then ξ_c is algebraic, (iv) gives $\Sigma(\xi) = S$, and $\gamma = \xi_{s_0}$ is algebraic. This is (ii). □

Proposition IV.6 (The Weil classes)**.** *For every CM field F, every $n \geq 2$ and every discriminant, the algebraic locus of the Weil classes in the polarised period domain $\mathcal{D}_F$ contains an explicit point of CM type and is dense, and it is either $\mathcal{D}_F$ or meagre.*

Proof. This is [7, Thm. 1.1], items (i) to (iii). The density is the Hecke orbit of the base point: the group $G_F(\mathbb{Q})$ of the unitary group of the hermitian form acts on $\mathcal{D}_F$ through F-linear isogenies that preserve the polarisation up to a scalar, hence preserve the algebraicity of the Weil classes, and its closure contains $G_F(\mathbb{R})^+$, which acts transitively. The dichotomy is Lemma IV.2. □

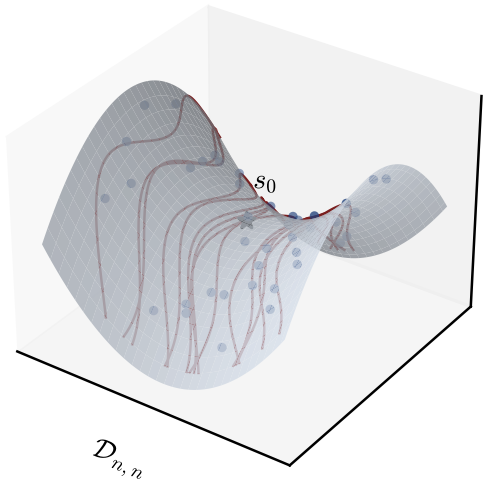


FIG. 2. The period domain $\mathcal{D}_{n,n}$ of a Weil family as a surface. The star is the CM base point, the dots and curves are strata of the algebraic locus, dense by the Hecke orbit. The locus is a countable union of closed strata; it is either the whole surface or meagre.

Proposition IV.7 ((F3′) and the abelian part)**.** *Let $H^{2p}_{\mathrm{ab}}(X) \subseteq H^{2p}(X)$ be the sum of the images of all maps $\Gamma_*\colon H^k(B) \to H^{2p}(X)$ with B abelian and Γ an algebraic cycle on $B \times X$. Then $H^{2p}_{\mathrm{ab}}(X)$ is a sub-Hodge structure, $\mathrm{Ab}^p(X) = \mathrm{Hdg}^p(X) \cap H^{2p}_{\mathrm{ab}}(X)$, and (F3′) holds for X in degree $2p$ if and only if a Hodge complement $T^{2p}_{\mathrm{ab}}(X)$ of $H^{2p}_{\mathrm{ab}}(X)$ carries no nonzero Hodge class.*

Proof. Each Γ_* is a morphism of Hodge structures, so its image and the sum of finitely many images, which is all of $H^{2p}_{\mathrm{ab}}(X)$ by dimension, are sub-Hodge structures. A Hodge class in $H^{2p}_{\mathrm{ab}}(X)$ is $\sum \Gamma_{j*} y_j$, and the argument of Proposition III.3(i) replaces each y_j by a Hodge class; so $\mathrm{Ab}^p(X) = \mathrm{Hdg}^p(X) \cap H^{2p}_{\mathrm{ab}}(X)$. A Hodge complement exists because the Hodge group is reductive, and the Hodge classes of $H^{2p}(X)$ are the direct sum of those of the two summands, which gives the last assertion. □

Theorem I.2 is Lemma IV.2, Theorem IV.5, Proposition IV.6 and Proposition IV.7. Figure 2 draws the locus of Proposition IV.6.

Both statements are now of one shape. There is a dense set of points at which the class is algebraic by an explicit cycle; the locus is a countable union of closed strata; the statement is that this union is not meagre, that is, that algebraicity propagates from the dense set. The rest of the paper goes to the first point where the propagation is not known and builds the object that would carry it.

V. CONSTRUCTIONS AT (F3′): REAL MULTIPLICATION ON K3 SURFACES

A. Where the first class is

Let S be a projective K3 surface, $T = T(S)_{\mathbb{Q}}$ and $E = \mathrm{End}_{\mathrm{Hdg}} T$. The Hodge classes of $S \times S$ are $H^0 \otimes H^4$, $H^4 \otimes H^0$ and $\mathrm{End}_{\mathrm{Hdg}} H^2(S) = \mathrm{End}(\mathrm{NS}_{\mathbb{Q}}) \oplus E$, since T is irreducible with no class of type $(1,1)$ and $\mathrm{Hom}_{\mathrm{Hdg}}(T, \mathrm{NS}_{\mathbb{Q}}) = 0$. The first two pieces and $\mathrm{End}(\mathrm{NS}_{\mathbb{Q}})$ are spanned by products of divisor classes. So the Hodge conjecture for $S \times S$ is the algebraicity of E, and by Proposition IV.7 the same classes decide (F3′) for $S \times S$ once T is not reached from abelian varieties. When E is a CM field it is spanned by its elements of norm one, which are Hodge isometries of T and algebraic by Buskin's theorem [8]; see [40, Rem. 2.16]. When E is totally real, the Rosati involution is the identity on it [46], every $e \in E$ is self-adjoint for the intersection form q, and $q(ex, ey) = q(e^2 x, y)$. The first class is a real multiplication.

B. Chains of correspondences

Lemma V.1. *Let $f\colon S \dashrightarrow S'$ be a dominant rational map of K3 surfaces of degree d and Γ_f the closure of its graph. Then $f_* = \Gamma_{f*}$ and $f^* = \Gamma_{f*}^t$ are morphisms of Hodge structures, f^* maps $T(S')_{\mathbb{Q}}$ isomorphically onto $T(S)_{\mathbb{Q}}$ with $q_S(f^*a, f^*b) = d\, q_{S'}(a,b)$, and $f_* = d\,(f^*)^{-1}$ on $T(S)_{\mathbb{Q}}$.*

Proof. The graph is an algebraic cycle, so both maps are induced by algebraic classes and are morphisms of Hodge structures. Pull-back of a nowhere vanishing 2-form is a nonzero 2-form, so $f^*(T(S')_{\mathbb{Q}})$ is a sub-Hodge structure of $H^2(S)$ that meets $H^{2,0}(S)$; the smallest sub-Hodge structure of $H^2(S)$ containing $H^{2,0}$ is $T(S)_{\mathbb{Q}}$, so $f^*T(S')_{\mathbb{Q}} \supseteq T(S)_{\mathbb{Q}}$, and since $T(S')_{\mathbb{Q}}$ is irreducible and f^* is injective on it ($f_* f^* = d$), equality of dimensions gives $f^*T(S')_{\mathbb{Q}} = T(S)_{\mathbb{Q}}$. The projection formula gives $q_S(f^*a, f^*b) = q_{S'}(f_* f^* a, b) = d\, q_{S'}(a,b)$ and $f_* f^* = d$. □

Lemma V.2. *A Hodge isometry $\varphi\colon T(S)_{\mathbb{Q}} \to T(S')_{\mathbb{Q}}$ is induced by an algebraic class on $S \times S'$.*

Proof. The spaces $\mathrm{NS}(S)_{\mathbb{Q}}$ and $\mathrm{NS}(S')_{\mathbb{Q}}$ are the orthogonal complements of $T(S)_{\mathbb{Q}}$ and $T(S')_{\mathbb{Q}}$ in the two copies of $\Lambda_{\mathbb{Q}}$, isometric to each other through φ; by Witt's cancellation theorem the complements are isometric, so φ extends to a Hodge isometry $H^2(S) \to H^2(S')$, which is algebraic by Buskin's theorem [8] (see [19, Thm. 0.1]). The projector of $H^2(S)$ onto $T(S)_{\mathbb{Q}}$ along $\mathrm{NS}(S)_{\mathbb{Q}}$ is algebraic, being 1 minus the projector onto NS, which is a sum $\sum_{i,j} c_{ij}\, D_i \otimes D_j$ over a basis of divisor classes with (c_{ij}) the inverse Gram matrix; composing gives φ on T and zero on NS. □

Theorem V.3 (Chains). *Let S be a projective K3 surface with E totally real and $e \in E$ with $e^2 = n \in \mathbb{Q}_{>0}$. Suppose there are projective K3 surfaces $S = S_0, S_1, \dots, S_r$ and, for each i, either a dominant rational map $S_{i-1} \dashrightarrow S_i$ of degree n_i, a dominant rational map $S_i \dashrightarrow S_{i-1}$ of degree n_i, or a Hodge isometry $T(S_{i-1})_{\mathbb{Q}} \cong T(S_i)_{\mathbb{Q}}$ (and then $n_i = 1$), such that*

$$\prod_i n_i^{\pm 1} \equiv n \pmod{\mathbb{Q}^{\times 2}},$$

the sign being $+$ for maps leaving S_{i-1} and $-$ for maps arriving at it. Then e is algebraic on $S \times S$.

Proof. Composing along the chain the maps f_* for maps leaving S_{i-1}, f^* for maps arriving, and the classes of Lemma V.2 for the isometries, gives an algebraic isomorphism $\phi\colon T(S_r)_{\mathbb{Q}} \to T(S)_{\mathbb{Q}}$ of Hodge structures with $q_S(\phi a, \phi b) = \mu\, q_{S_r}(a,b)$, where $\mu = \prod n_i^{\pm 1}$ by Lemma V.1 (a map leaving contributes $f_* = n_i (f^*)^{-1}$, of multiplier n_i; a map arriving contributes f^*, of multiplier n_i in the other direction, that is n_i^{-1} for the composite in the direction of ϕ). Write $\mu = nc^2$ with $c \in \mathbb{Q}^{\times}$. Since e is self-adjoint with $e^2 = n$, it is a Hodge similarity of $T(S)_{\mathbb{Q}}$ of multiplier n. So $\psi = (c^{-1}\phi)^{-1} \circ e\colon T(S)_{\mathbb{Q}} \to T(S_r)_{\mathbb{Q}}$ is a Hodge isometry, algebraic by Lemma V.2, and $e = (c^{-1}\phi) \circ \psi$ is a composite of algebraic correspondences, hence algebraic on $S \times S$. □

The case $r = 1$ with $S \dashrightarrow S_1$ the quotient by a symplectic automorphism of prime order p is Varesco's theorem [40, Thm. 2.1], which gives $e = \sqrt{p}$ and is used for $p = 2, 3$. What a chain adds is a composite multiplier.

C. Real multiplication by $\mathbb{Q}(\sqrt{6})$

A symplectic automorphism of finite order on a K3 surface acts trivially on $T(S)$, its quotient has a K3 resolution Y [33], and $T(Y)_{\mathbb{Q}} \cong T(S)_{\mathbb{Q}}(1/\mathrm{ord})$ by Lemma V.1. The invariant lattice of a symplectic involution is $U^3 \oplus E_8(-2)$, and a K3 surface has a symplectic involution exactly when $E_8(-2)$ embeds primitively in its Néron–Severi lattice [29, 43]. The invariant lattice of a symplectic automorphism of order three is $U \oplus U(3)^2 \oplus A_2^2$, and such an automorphism exists exactly when the Coxeter–Todd lattice $K_{12}(-2)$ embeds primitively in the Néron–Severi lattice [16, Thm. 4.1]. Put

$$V_2 = \big(U^3 \oplus E_8(-2)\big)_{\mathbb{Q}}, \qquad V_3 = \big(U \oplus U(3)^2 \oplus A_2^2\big)_{\mathbb{Q}},$$

of ranks 14 and 10 and signatures $(3,11)$ and $(3,7)$; over $\mathbb{Q}$, $U(3) \cong U$ and $E_8(-2) \cong \langle -1\rangle^8 \cong \langle -2\rangle^8$. Varesco [40, Props. 2.5 and 2.11] shows that a projective K3 surface whose transcendental space embeds isometrically in V_2 (resp. V_3) is Hodge isometric to one with a symplectic involution (resp. a symplectic automorphism of order three).

Lemma V.4 (Embedding criterion). *Let W and V be rational quadratic spaces with $\operatorname{rk} V - \operatorname{rk} W \geq 3$. Then W embeds isometrically in V if and only if the signature of V dominates that of W in both signs. In that case a complement W' exists with $V \cong W \oplus W'$.*

Proof. Necessity is clear. For sufficiency put $k = \operatorname{rk} V - \operatorname{rk} W \geq 3$ and let (s_+, s_-) be the difference of signatures, $s_+ + s_- = k$. A quadratic space W' of rank k and signature (s_+, s_-) with prescribed discriminant $d' = d(V)d(W)$ and prescribed Hasse invariants ε'_p exists provided the ε'_p equal 1 for almost all p, $\prod_{p \leq \infty} \varepsilon'_p = 1$, and ε'_∞, d' match the signature [36, Chap. IV, Prop. 7], with no further condition when $k \geq 3$. Take $\varepsilon'_p = \varepsilon_p(V)\varepsilon_p(W)(d(W), d')_p$, the value forced by the formula $\varepsilon_p(W \oplus W') = \varepsilon_p(W)\varepsilon_p(W')(d(W), d(W'))_p$. The product formula for these holds because it holds for V, for W and for the Hilbert symbol. Then $W \oplus W'$ and V have the same rank, signature, discriminant and Hasse invariants at every place, so they are isometric by the Hasse–Minkowski theorem [36, Chap. IV, Thm. 9], and W embeds in V. □

Theorem V.5 (Two-dimensional families with algebraic real multiplication by $\mathbb{Q}(\sqrt{6})$). *Let S be a projective K3 surface with $\mathbb{Q}(\sqrt{6}) \subseteq E$ whose transcendental space $T = T(S)_{\mathbb{Q}}$ embeds isometrically in V_3.*

(i) *$\sqrt{6} \in E$ is algebraic on $S \times S$.*

(ii) *The Hodge conjecture, and* (F3′)*, hold for $S \times S$ and for the Hilbert square $S^{[2]}$.*

(iii) *Such surfaces exist in two-dimensional families, whose very general member has $\operatorname{rk} T = 8$ and $E = \mathbb{Q}(\sqrt{6})$ exactly. One such family is the one whose transcendental space is E^4 with the trace form of $q_E = \langle -2, -2, -2+\sqrt{6}, -1+\sqrt{6}\rangle$ over $E = \mathbb{Q}(\sqrt{6})$.*

Proof. (i) By [40, Prop. 2.11] S is Hodge isometric to a K3 surface S' with a symplectic automorphism σ of order three, and by Lemma V.2 the isometry is algebraic, so we may replace S by S'. Let Y be the minimal resolution of S'/σ and $f_1\colon S' \dashrightarrow Y$ the quotient map, of degree three, so that $T(Y)_{\mathbb{Q}} \cong T(1/3)$, of rank $\operatorname{rk} T \leq 8$ and signature $(2, \operatorname{rk} T - 2)$: the rank is at most 8 because T embeds in V_3 of rank 10 and signature $(3,7)$ with a complement that must contain the remaining positive direction, so the complement has rank at least two. Now $T(Y)_{\mathbb{Q}}$ has rank at most 8 and signature dominated by $(3,11)$, and $14 - 8 \geq 3$, so by Lemma V.4 it embeds in V_2. By [40, Prop. 2.5] Y is Hodge isometric to a K3 surface Y' with a symplectic involution ι; let Z be the resolution of Y'/ι and $f_2\colon Y' \dashrightarrow Z$ the quotient, of degree two. The chain $S' \dashrightarrow Y \cong Y' \dashrightarrow Z$ has multiplier $3 \cdot 1 \cdot 2 = 6$, and Theorem V.3 applies to $e = \sqrt{6}$ (Fig. 3).

(ii) By the first paragraph of this section it remains to make E algebraic. E contains $\mathbb{Q}(\sqrt{6})$, and either $E = \mathbb{Q}(\sqrt{6})$ or E is a CM field: a totally real E has $\dim_E T \geq 3$, because the complex line $H^{2,0}$ lies in one eigenspace

TABLE II. Rational invariants used in Theorem V.5(iii). Hasse invariants are $+1$ at every prime not shown.

space	rank	signature	d	ε_2	ε_3	ε_5
$T=\mathrm{Tr}_{E/\mathbb{Q}}q_E$	8	$(2,6)$	10	$+1$	-1	$+1$
$\langle 1,-10\rangle$	2	$(1,1)$	-10	$+1$	$+1$	$+1$
$T\oplus\langle 1,-10\rangle$	10	$(3,7)$	-1	$+1$	-1	$+1$
V_3	10	$(3,7)$	-1	$+1$	-1	$+1$
V_2	14	$(3,11)$	-1	-1	$+1$	$+1$

$T_\sigma\otimes\mathbb{C}$ and $\overline{H^{2,0}}$ in the same one, so that $T_\sigma\otimes\mathbb{R}$ has signature $(2,*)$ and $\dim_E T\geq 3$ follows from q being negative definite on the other eigenspaces together with $T\otimes\mathbb{R}$ having exactly two positive directions; and $[E:\mathbb{Q}]\cdot\dim_E T=\mathrm{rk}\,T\leq 8$ then forces $[E:\mathbb{Q}]\leq 2$. If $E=\mathbb{Q}(\sqrt{6})$, (i) suffices. If E is CM, it is spanned by its norm-one elements, Hodge isometries, algebraic by Buskin's theorem. So every Hodge class on $S\times S$ is algebraic, and (F3′) follows. For $S^{[2]}$ the Chow motive is a direct summand of twists of $h(S\times S)$, $h(S)$ and $\mathbf{1}$ [10], so every Hodge class on $S^{[2]}$ is the image under an algebraic correspondence of a Hodge class on $S\times S$ or S, and both statements follow.

(iii) Let $E=\mathbb{Q}(\sqrt{6})$ with real embeddings σ_1, σ_2 ($\sqrt{6}\mapsto\pm\sqrt{6}$), and let $q_E=\langle a_1,a_2,a_3,a_4\rangle$ with a_1,a_2 negative at both places and a_3,a_4 positive at σ_1, negative at σ_2. The $\mathbb{Q}$-space $T=E^4$ with the form $\mathrm{Tr}_{E/\mathbb{Q}}\circ q_E$ has rank 8 and signature $(2,2)+(0,4)=(2,6)$, and multiplication by $\sqrt{6}$ is self-adjoint of square 6. For $q_E=\langle -2,-2,-2+\sqrt{6},-1+\sqrt{6}\rangle$ one computes $d(T)=10$, $\varepsilon_2=1$, $\varepsilon_3=-1$, $\varepsilon_5=1$ and $\varepsilon_p=1$ otherwise, while V_3 has $d=-1$, $\varepsilon_2=1$, $\varepsilon_3=-1$, $\varepsilon_5=1$; and $T\oplus\langle 1,-10\rangle$ has the invariants of V_3 at every place, so $V_3\cong T\oplus\langle 1,-10\rangle$ by Hasse–Minkowski, with a complement containing a positive vector (Table II; the Hilbert symbols are recomputed in the programs of Appendix A). Embed V_3 in $\Lambda_{\mathbb{Q}}$ as the invariant lattice of an order-three symplectic action, with orthogonal complement $K_{12}(-2)$, and let $N\subseteq\Lambda$ be the primitive sublattice with $N_{\mathbb{Q}}=T^\perp$; it contains $K_{12}(-2)$ primitively and a positive class. The Hodge structures of K3 type on T for which $\sqrt{6}$ is a Hodge endomorphism are the periods $\omega\in\mathbb{P}(T_{\sigma_1}\otimes\mathbb{C})$ with $q(\omega)=0$, $q(\omega,\bar\omega)>0$, where T_{σ_1} is the σ_1-eigenspace, of signature $(2,2)$; this is a domain of dimension two [40, Prop. 2.3]. For each such ω the surjectivity of the period map gives a K3 surface S_ω with $T(S_\omega)_{\mathbb{Q}}\subseteq T$, projective because N contains a positive class, carrying an order-three symplectic automorphism because $K_{12}(-2)\subseteq N$ primitively. For ω outside a countable union of proper closed analytic subsets, $T(S_\omega)_{\mathbb{Q}}=T$ and $E=\mathbb{Q}(\sqrt{6})$, by the argument of [7, Thm. 22.9], which needs only $\dim_E T\geq 3$. □

Varesco's families have $E=\mathbb{Q}(\sqrt{2})$ or $\mathbb{Q}(\sqrt{3})$ [40, Thms. 2.9, 2.15]; van Geemen and Schütt obtain $\sqrt{2}$, $\sqrt{3}$, $\sqrt{7}$ from rational self-maps of elliptic K3 surfaces [44, §§6–7]; Bayer-Fluckiger, van Geemen and Schütt prove that families with real multiplication by any totally real E with $[E:\mathbb{Q}]\dim_E T\leq 21$ exist, without algebraicity [9]. The lattice condition of Theorem V.5 is a genuine restriction: among the 165 sign-admissible diagonal forms q_E with small coefficients, 98 embed in V_3 and 67 do not (Appendix A).

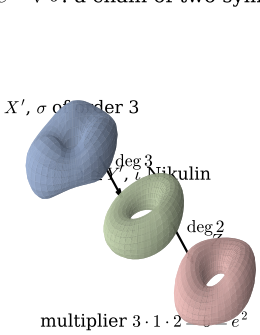


FIG. 3. The chain of Theorem V.5: the quotient of S' by a symplectic automorphism of order three, a Hodge isometry (algebraic by Buskin), and the quotient of Y' by a Nikulin involution. The multipliers compose to 6.

D. Isogenies of elliptic surfaces

Varesco's quotients and the chains of Theorem V.3 produce the square roots of the orders of symplectic automorphisms, 2 and 3, and their products. A different source of similarities is a rational self-map of a K3 surface, and the self-maps that exist in families come from isogenies of elliptic fibrations: van Geemen and Schütt obtain $\sqrt{2}$ and $\sqrt{3}$ from isogenies of degree 2 and 3 in this way, and for degrees 5 and 7 they give one surface each [44, §§6–7]. We show that the degrees 5 and 7 also come in families.

Let $N\in\{5,7\}$ and write N^k for 125 when $N=5$ and 49 when $N=7$. The modular curve $X_0(N)$ is rational, with Fricke's coordinate t in which the involution $(E,C)\mapsto(E/C,E[N]/C)$ is $w(t)=N^k/t$ [44, §7.6]. The universal curve with a cyclic subgroup C_t of order N over $\mathbb{P}^1_t$ is

$$\mathcal{E}_t:\ y^2=x^3-27c_4(t)\,x-54c_6(t),\tag{1}$$

with, for $N=7$,

$$c_4=(t^2+13t+49)(t^2+5t+1),$$
$$c_6=(t^2+13t+49)(t^4+14t^3+63t^2+70t-7),$$

and for $N=5$,

$$c_4=(t^2+10t+5)(t^2+22t+125),$$
$$c_6=(t^2+4t-1)(t^2+22t+125)^2.$$

The discriminant is $1728\,t\,(t^2+13t+49)^2$ for $N=7$ and $1728\,t\,(t^2+22t+125)^3$ for $N=5$, of degree 5 and 7. So the relatively minimal model R_N of (1) over $\mathbb{P}^1_t$ is a rational elliptic surface with fibres of type I_1 at $t=0$, I_N at $t=\infty$, and at the two roots e_1, e_2 of the quadratic factor, where c_4 vanishes, of type II for $N=7$ and III for $N=5$; the Euler numbers add up to $1+N+2\cdot 2=12$ and $1+N+2\cdot 3=12$. The roots satisfy $e_1e_2=N^k$, so w exchanges them and exchanges 0 with ∞.

Lemma V.6 (The Fricke isogeny over the function field)**.** *$\mathcal{E}_t/C_t \cong \mathcal{E}_{w(t)}$ over $\mathbb{C}(t)$.*

Proof. The two curves have the same j-invariant, because w is the Fricke involution of $X_0(N)$. Two elliptic curves over $\mathbb{C}(t)$ with the same non-constant j-invariant differ by a quadratic twist $\chi \in \mathbb{C}(t)^\times/\mathbb{C}(t)^{\times 2}$, and a twist ramified at a point of $\mathbb{P}^1$ changes the Kodaira type of the fibre there, exchanging I_n with I_n^*, II with IV* and III with III*. So it suffices to show that $\mathcal{E}_t/C_t$ has the same fibre types as $\mathcal{E}_{w(t)}$, which are I_N at 0, I_1 at ∞ and II or III at e_1, e_2: then χ is unramified on $\mathbb{P}^1$, hence constant, hence trivial over $\mathbb{C}$.

At $t=0$ the fibre of $\mathcal{E}$ is I_1, the Tate curve $\mathbb{G}_m/q^{\mathbb{Z}}$ over $\mathbb{C}((t))$, whose only cyclic subgroup of order N rational over $\mathbb{C}((t))$ is μ_N; the quotient $\mathbb{G}_m/q^{N\mathbb{Z}}$ is I_N. At e_i the local monodromy $M \in \mathrm{SL}_2(\mathbb{Z})$ of $\mathcal{E}$ has order 6 or 4 and irreducible characteristic polynomial; the isogeny is given on the period lattices by a matrix $A \in M_2(\mathbb{Z})$ with $\det A = N$, and the monodromy of the quotient is $M' = AMA^{-1}$. The $\mathrm{SL}_2(\mathbb{Z})$-classes with this characteristic polynomial are those of M and M^{-1}, which are the types II and II*, or III and III*. If $M' = gM^{-1}g^{-1}$ with $g \in \mathrm{SL}_2(\mathbb{Z})$, then $B = g^{-1}A$ conjugates M to M^{-1}. All rational matrices doing so form the coset $B_0\,\mathbb{Q}(M)^\times$, where $B_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ for $M = \left(\begin{smallmatrix} 0 & -1 \\ 1 & \mathrm{tr}M \end{smallmatrix}\right)$ has determinant -1 and the elements of the imaginary quadratic field $\mathbb{Q}(M)$ have positive norm; so $\det B < 0$, against $\det B = N$. Hence the quotient has the type of M at e_1, e_2. At ∞ the j-invariant of the quotient has a simple pole, so the fibre is I_1 or I_1^*, and the Euler numbers of the other fibres add up to 11; since the total is 12 the fibre is I_1. □

For $p \neq q$ in $\mathbb{P}^1_t$ let $u_{p,q} = (t-p)(t-q)$, and let $X_{p,q}$ be the elliptic K3 surface whose generic fibre is the quadratic twist $\mathcal{E}_t \otimes u_{p,q}$: the twist multiplies c_4 by u^2 and c_6 by u^3, giving Weierstrass coefficients of degrees 8 and 12, and adds two fibres of type I_0^* at p and q when these are not among $0, \infty, e_1, e_2$. For a double cover $\psi\colon \mathbb{P}^1_s \to \mathbb{P}^1_t$ let X_ψ be the relatively minimal model of $\psi^*\mathcal{E}$.

Lemma V.7.

(i) *$X_{p,q}$ has Picard number at least 16: its Néron–Severi lattice contains $U \oplus A_6 \oplus D_4^2$ for $N=7$ and $U \oplus A_4 \oplus A_1^2 \oplus D_4^2$ for $N=5$. The same bound holds for X_ψ, whose fibres $2\,\mathrm{I}_1 + 2\,\mathrm{I}_N$, together with four fibres of type* III *when $N=5$, and the pulled-back Mordell–Weil group of R_N, of rank 2, give rank at least 16 by the Shioda–Tate formula.*

(ii) *The fibrewise isogeny by C_t is a dominant rational map of degree N*

$$f\colon X_{p,q} \dashrightarrow X_{w(p),w(q)}, \qquad f\colon X_\psi \dashrightarrow X_{w\circ\psi}.$$

(iii) *The deck involution of ψ times $[-1]$ on the fibres is a symplectic involution ι of X_ψ, and the minimal resolution of X_ψ/ι is $X_{p,q}$ with $\{p,q\}$ the branch locus of ψ. Hence $T(X_\psi)_\mathbb{Q} \cong T(X_{p,q})_\mathbb{Q}(1/2)$ as in Lemma V.1, and the two surfaces have the same Picard number.*

Proof. (i) Shioda–Tate with the fibres listed, and for X_ψ the Mordell–Weil rank $8-6=2$ of R_N. (ii) By Lemma V.6 the quotient of the generic fibre is $\mathcal{E}_{w(t)} \otimes u_{p,q}(t)$, and

$$u_{p,q}(w(t)) = \Big(\frac{N^k}{t} - p\Big)\Big(\frac{N^k}{t} - q\Big) = \frac{pq}{t^2}\,u_{w(p),w(q)}(t),$$

a constant times a square times $u_{w(p),w(q)}$. So $\mathcal{E}_{w(t)} \otimes u_{p,q}(t) \cong \mathcal{E}_{w(t)} \otimes u_{w(p),w(q)}(w(t))$, which is the pull-back along the automorphism w of the base of the generic fibre of $X_{w(p),w(q)}$; pulling back along an automorphism of the base gives an isomorphic surface. For X_ψ, $(\psi^*\mathcal{E})/C = \psi^*(\mathcal{E}/C) \cong \psi^*w^*\mathcal{E}$. (iii) The quadratic twist by the function defining the double cover is the quotient of the base change by the deck involution composed with $[-1]$ on the fibres; the involution is symplectic with eight fixed points, the 2-torsion of the two fibres over the ramification points, and the quotient's resolution is the twist. □

Theorem V.8 (One-dimensional families with algebraic real multiplication by $\sqrt{5}$ and $\sqrt{7}$)**.** *Let $N \in \{5,7\}$ and let $\mathfrak{A}_N$ be the family of surfaces $X_p = X_{p,N^k/p}$ for $p \notin \{0, \infty, e_1, e_2, \pm\sqrt{N^k}\}$, and $\mathfrak{B}_N$ the family of surfaces X_ψ with ψ branched over a pair $\{b, N^k/b\}$.*

(i) *Every member X has a dominant rational self-map f of degree N, and f^* is an element of $E = \mathrm{End}_{\mathrm{Hdg}}T(X)_\mathbb{Q}$ with $f^*f_* = N$.*

(ii) *The families are not isotrivial. Their very general member has Picard number 16, transcendental rank 6 and $E = \mathbb{Q}(\sqrt{N})$, with $\sqrt{N} = \pm f^*$ algebraic on $X \times X$.*

(iii) *For every member X the Hodge conjecture holds for $X \times X$, and* (F3′) *holds for $X \times X$ and for $X^{[2]}$.*

Proof. (i) w exchanges p and N^k/p, so Lemma V.7(ii) gives $f\colon X_p \dashrightarrow X_p$; for $\mathfrak{B}_N$, $w \circ \psi$ has the same branch locus as ψ, so $w \circ \psi = \psi \circ \sigma$ for a Möbius transformation σ and $X_{w\circ\psi} \cong X_\psi$. The correspondence of a dominant rational map preserves NS and its transpose does too, so it preserves $T = \mathrm{NS}^\perp$ and acts there as a morphism of Hodge structures, nonzero since $f^*\omega = \lambda\omega$ with $\lambda \neq 0$; on a resolution S of the graph with $p_1\colon S \to X$ birational and p_2 of degree N, both p_1^*T and p_2^*T are irreducible sub-Hodge structures of $H^2(S)$ containing its unique $(2,0)$-line, so they coincide, and $f_*f^* = p_{2*}p_1^*p_{1*}p_2^* = N$ on T.

(ii) The elliptic fibration of X_p has labelled singular fibres I_1 at 0, I_N at ∞, II or III at e_1, e_2 and I_0^* at p and N^k/p. An isomorphism of elliptic fibrations induces a Möbius transformation preserving this configuration; the transformations preserving $\{0,\infty\}$ and $\{e_1,e_2\}$ are $t \mapsto \lambda t$ and $t \mapsto \lambda/t$ with $\lambda e_1 \in \{e_1,e_2\}$, and since $e_1+e_2 \neq 0$ and $e_1^2 \neq e_2^2$ they are the identity and w, which preserve $\{p, N^k/p\}$. So X_p and $X_{p'}$ have isomorphic fibrations only when $p' \in \{p, N^k/p\}$. A K3 surface has finitely many elliptic fibrations up to automorphisms [39], so the map $p \mapsto [X_p]$ to the moduli space has finite fibres and the family is not isotrivial; by local Torelli its period map has uncountable image. A Hodge structure of K3 type on a fixed rational quadratic space with E a CM field is determined by E, a subalgebra of the endomorphism algebra, and by an eigenline of $E\otimes\mathbb{C}$, so there are countably many; the very general member is therefore not of CM type, and its E is totally real. It contains f^*, and $f_* = N(f^*)^{-1}$ is the adjoint of f^*, equal to f^* since the adjoint involution is trivial on a totally real E; so $(f^*)^2 = N$ and $E \neq \mathbb{Q}$. A totally real $E \neq \mathbb{Q}$ has $\dim_E T \geq 3$ [42], so $\mathrm{rk}\, T \geq 6$; with $\rho \geq 16$ this gives $\mathrm{rk}\, T = 6$, $\rho = 16$, $[E:\mathbb{Q}] = 2$ and $E = \mathbb{Q}(\sqrt{N})$. The same holds for $\mathfrak{B}_N$ by Lemma V.7(iii).

(iii) For every member $E \supseteq \mathbb{Q}(\sqrt{N})$ and $\mathrm{rk}\, T \leq 6$. If E is totally real then $\dim_E T \geq 3$ forces $E = \mathbb{Q}(\sqrt{N}) = \mathbb{Q} + \mathbb{Q}f^*$, algebraic. Otherwise E is a CM field, and every element of E of norm one to its real subfield is a Hodge isometry of T, extended by the identity on NS to a Hodge isometry of $H^2(X,\mathbb{Q})$, algebraic by Buskin's theorem [8]. The algebraic classes form a $\mathbb{Q}$-subalgebra $E' \subseteq E$ containing these elements; for $x \in E$ with $\bar{x} \neq x$ and $t_1 \neq t_2$ in $\mathbb{Q}$ it contains $(t_i + x)/(t_i + \bar{x})$, hence their quotient $1+(t_2-t_1)/(t_1+\bar{x})$, hence $\bar{x}$; so $E' = E$. In both cases $\mathrm{End}_{\mathrm{Hdg}} H^2(X) = \mathrm{End}(\mathrm{NS}) \oplus E$ is algebraic, and $\mathrm{Hdg}^2(X\times X) = \mathrm{End}_{\mathrm{Hdg}} H^2(X) \oplus H^0\otimes H^4 \oplus H^4 \otimes H^0$ since $H^1 = H^3 = 0$; the Hodge conjecture holds for $X \times X$, which gives (F3′) there, and passes to $X^{[2]}$ by [10]. □

The surface of [44, §7.6] with real multiplication by $\sqrt{7}$ is the twist at $\{7,-7\}$, the two fixed points of w, and the surface of [44, Prop. 7.7] is the twist at the fixed points $\pm 5\sqrt{5}$ for $N = 5$, which has Picard number 18 and complex multiplication. The families $\mathfrak{A}_N$ are the twists at the pairs exchanged by w; they are one-dimensional, their very general member has real and not complex multiplication, and for $N = 5$ they are the first K3 surfaces of transcendental rank six with real multiplication by $\sqrt{5}$ that we know to carry an algebraic $\sqrt{5}$. The chains of Theorem V.3 apply to every member: a transcendental space of rank six embeds in V_2 and V_3 with any rational scaling, so a member whose E happened to contain $\sqrt{2N}$, $\sqrt{3N}$ or $\sqrt{6N}$ would have that element algebraic as well. We do not know such a member: the twist families of fixed Picard number 16 are two-dimensional, the real multiplication loci for a fixed quadratic field on a transcendental space of rank six are one-dimensional inside a four-dimensional period domain, and the intersection has expected dimension -1.

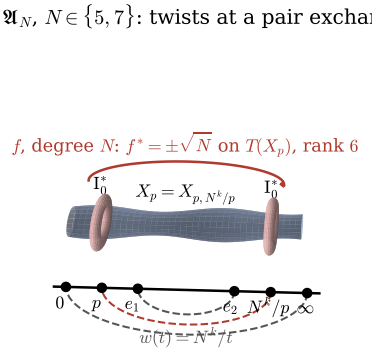


FIG. 4. The family $\mathfrak{A}_N$ of Theorem V.8. Over the Fricke coordinate t of $X_0(N)$ the involution $w(t) = N^k/t$ exchanges the cusps 0, ∞ and the elliptic points e_1, e_2; the twist X_p at a pair $\{p, N^k/p\}$ exchanged by w is carried to itself by the fibrewise isogeny f of degree N, and f^* is the real multiplication by $\sqrt{N}$ on the transcendental lattice of rank six.

E. The reach of similarities

Every correspondence used so far acts on transcendental spaces as a Hodge similarity. The next proposition says that this is not an accident of the examples but a property of every correspondence whose components are small, and it measures what a correspondence must look like to act otherwise.

Proposition V.9 (Structure of the transcendental action)**.** *Let X, X' be projective K3 surfaces, $\Gamma \in \mathrm{CH}^2(X\times X')_{\mathbb{Q}}$, and $\gamma_T \colon T(X)_{\mathbb{Q}} \to T(X')_{\mathbb{Q}}$ the map induced on transcendental spaces by composing the action of Γ with the orthogonal projections.*

(i) *γ_T is zero or an isomorphism of Hodge structures.*

(ii) *If $\gamma_T \neq 0$ then $a = \gamma_T^\dagger \gamma_T$ lies in the maximal totally real subfield of $E = \mathrm{End}_{\mathrm{Hdg}} T(X)_{\mathbb{Q}}$ and is totally positive, and γ_T is a similarity exactly when $a \in \mathbb{Q}$.*

(iii) *Let Γ be irreducible, S a resolution of its normalisation with projections p_1, p_2 of degrees d_1, d_2 to X, X'. If $p_1^* T(X)_{\mathbb{Q}} = p_2^* T(X')_{\mathbb{Q}}$ inside $H^2(S,\mathbb{Q})$, then γ_T is a similarity of multiplier $d_1 d_2$. This holds whenever $p_g(S) \leq 1$.*

(iv) *If Γ is irreducible and γ_T is not a similarity, then $p_1^* T(X)_{\mathbb{Q}} \cap p_2^* T(X')_{\mathbb{Q}} = 0$, so $p_g(S) \geq 2$ and the transcendental rank of S is at least twice that of X.*

(v) *The $\mathbb{Q}$-algebra generated by the actions on $T(X)_{\mathbb{Q}}$ of all correspondences from X to itself whose irre-*

ducible components either fail to dominate a factor or have $p_g \le 1$ is the multiquadratic subfield $E_2 \subseteq E$ generated by the square roots of the products $d_1 d_2$ that occur. When $[E:\mathbb{Q}]$ is odd it is $\mathbb{Q}$.

Proof. (i) $T(X)_{\mathbb{Q}}$ is an irreducible Hodge structure [46], and the image of γ_T is a sub-Hodge structure of $H^2(X')$; if nonzero it contains the $(2,0)$-part, hence $T(X')_{\mathbb{Q}}$, and γ_T is injective by irreducibility. (ii) a is a Hodge endomorphism of $T(X)$, so $a \in E$, and it is self-adjoint, so it lies in the fixed field of the adjoint involution, the maximal totally real subfield. The form $q'(\gamma x, \gamma y) = q(x, ay)$ is a polarisation of the Hodge structure $T(X)$ up to sign on each real eigenspace of E, and comparison with q gives $\sigma(a) > 0$ at every real place. A similarity of multiplier μ has $a = \mu$. (iii) Both $p_i^* T$ are irreducible sub-Hodge structures of $H^2(S)$ with a $(2,0)$-part; if $p_g(S) \le 1$ they share the unique $(2,0)$-line, so they meet and coincide. When they coincide, $\gamma_T = p_{2*} p_1^* = d_2 (p_2^*)^{-1} p_1^*$ on T, so $p_2^* \gamma x = d_2\, p_1^* x$; since $q_S(p_1^* x, p_1^* y) = d_1 q(x,y)$ and $q_S(p_2^* x', p_2^* y') = d_2 q'(x', y')$, this gives $d_2\, q'(\gamma x, \gamma y) = q_S(p_2^* \gamma x, p_2^* \gamma y) = d_2^2\, q_S(p_1^* x, p_1^* y) = d_1 d_2^2\, q(x,y)$, that is $q'(\gamma x, \gamma y) = d_1 d_2\, q(x,y)$. (iv) If the two subspaces met they would coincide by irreducibility and (iii) would apply; so they are independent, their $(2,0)$-lines are distinct, and their sum has rank $2 \operatorname{rk} T(X)$ inside the transcendental part of $H^2(S)$. (v) A component that does not dominate a factor acts through NS or through H^0 and H^4, hence by zero on T. The others act by similarities by (iii), a composite of similarities returning to X is an element $\Phi \in E$ with $\Phi^\dagger \Phi \in \mathbb{Q}$, and for totally real E this is $\Phi^2 \in \mathbb{Q}$; such elements generate E_2. For a CM field E the elements with $\Phi^\dagger \Phi \in \mathbb{Q}$ are the $\mathbb{Q}^\times$-multiples of the norm-one elements, which generate E, so the statement is about the totally real case, which is the one that matters for (F3′). An odd-degree field has $E_2 = \mathbb{Q}$. □

The correspondences of Lemmas V.1 and V.2, the Fourier–Mukai kernels between a K3 surface and its moduli spaces of sheaves, the rational Hodge isometries of hyper-Kähler varieties of $\mathrm{K3}^{[n]}$ type [24], the Fricke isogenies of Theorem V.8 and all their composites fall under (iii) or (v). A correspondence that realises an element of a cubic field must therefore have a component whose resolution has $p_g \ge 2$ and transcendental rank at least twice that of the surface. Such correspondences exist, and the only ones we know come from finite groups.

Proposition V.10 (Hecke correspondences)**.** *Let Z be a smooth projective surface with an action of a finite group G, $H \le G$ a subgroup, and X a smooth projective surface with $p_g(X) = 1$ birational to Z/H; write $\pi\colon Z \dashrightarrow X$.*

(i) *For $g \in G$ the correspondence $\Gamma_g = (\pi \times \pi)_*[\text{graph of } g]$ acts on $H^2(X,\mathbb{Q})$ as the Hecke operator of the double coset HgH, and these operators span a $\mathbb{Q}$-algebra isomorphic to a quotient of $\mathrm{End}_G \mathbb{Q}[G/H]$ acting by Hodge endomorphisms.*

(ii) *Exactly one simple factor of this algebra acts non-trivially on $H^{2,0}(X)$; it is a field, it preserves $T(X)_{\mathbb{Q}}$, and it embeds into E by algebraic correspondences. If $H^{2,0}(Z)^G = 0$ this factor is not the one on which G acts trivially.*

(iii) *Every field obtained in this way is abelian over $\mathbb{Q}$. In particular a totally real cubic field F whose Galois closure has group S_3 is realised by no correspondence built from a finite group acting on a cover of X.*

Proof. (i) Let $\tilde{Z} \to Z$ be an equivariant resolution and $\pi\colon \tilde{Z} \to X$ the quotient by H followed by the resolution; then $\pi_* \pi^* = |H|$ and $\pi_* h^* = \pi_*$ for $h \in H$, so Γ_g acts as $\pi_* g^* \pi^* = (\pi^*)^{-1} \circ \sum_{h \in HgH/H} h^* \circ \pi^*$ on $H^2(X,\mathbb{Q}) = H^2(\tilde{Z},\mathbb{Q})^H$, the Hecke operator of HgH. The Hecke operators of (G,H) span $\mathrm{End}_G \mathbb{Q}[G/H]$, a semisimple algebra, and the action on $H^2(\tilde{Z},\mathbb{Q})^H$ factors through it. (ii) The idempotents of the simple factors are rational combinations of Hecke operators, so they are Hodge endomorphisms of $H^2(X)$; they sum to 1 and act on the line $H^{2,0}(X)$ by 0 or 1, so exactly one of them, e_V, acts as 1. A simple factor with nonzero image in the commutative ring $\mathrm{End}_{\mathrm{Hdg}} H^{2,0}(X) = \mathbb{C}$ through a ring homomorphism to a field is commutative, hence a field $\mathcal{H}_V$. For $f \in \mathcal{H}_V$, $f(T(X)_{\mathbb{Q}})$ is a sub-Hodge structure of the transcendental part of $H^2(X)$; if $f\omega \ne 0$ it contains ω, hence $T(X)_{\mathbb{Q}}$, and if $f\omega = 0$ it has no $(2,0)$-part and lies in $\mathrm{NS} \cap T = 0$. So $\mathcal{H}_V \to \mathrm{End}_{\mathrm{Hdg}} T(X)_{\mathbb{Q}} = E$ is a homomorphism from a field, injective. The factor on which G acts trivially fixes $H^{2,0}(\tilde{Z})^G = H^{2,0}(Z/G)$, which is zero by hypothesis, so it is not $\mathcal{H}_V$. (iii) The centres of the simple factors of $\mathrm{End}_G \mathbb{Q}[G/H]$ are the character fields $\mathbb{Q}(\chi_i)$ of the irreducible constituents of $\mathbb{Q}[G/H]$, generated by sums of roots of unity, so each lies in $\mathbb{Q}(\zeta_{|G|})$ and is abelian; a commutative simple factor is its centre. A non-abelian cubic field is not a subfield of an abelian field. □

For the dihedral group D_n with n odd, $H = \langle \tau \rangle$ of order two and Z the elliptic surfaces $\tilde{\mathcal{E}}_a$ of [44, §4.7], the algebra $\mathrm{End}_{D_n} \mathbb{Q}[D_n/H]$ is $\mathbb{Q} \times \prod_{d \mid n,\, d>1} \mathbb{Q}(\zeta_d)^+$, and the factor acting on ω is $\mathbb{Q}(\zeta_n)^+$; this is the content of [44, §4.8], where the cycle $\Gamma_1 + \Gamma_{-1}$ induces $\zeta_n + \zeta_n^{-1}$. The components of these correspondences are birational to $\tilde{\mathcal{E}}_a$, which has $p_g \ge 2$, as Proposition V.9(iv) requires.

Corollary V.11 (Squares of the dihedral families)**.** *Let X be an elliptic K3 surface of one of the families of [44, Thm. 1.1], with real multiplication by $F = \mathbb{Q}(\zeta_n)^+$ for $n \in \{5,7,9,11\}$, and suppose $E = F$ or E is a CM field; the very general member has $E = F$. Then the Hodge conjecture holds for $X \times X$, and (F3′) holds for $X \times X$ and $X^{[2]}$. This covers the cubic fields $\mathbb{Q}(\zeta_7)^+$ and $\mathbb{Q}(\zeta_9)^+$ on families of dimensions 3 and 2.*

Proof. If $E = F$, E is algebraic by Proposition V.10(ii) and [44, §4.8]; if E is CM, by Buskin's theorem as in Theorem V.8(iii). The rest is as there. □

F. Where this stops

The frontier case of (F3′) in [7, Thm. 21.4 and Rem. 21.5] is a K3 surface S_λ of Picard number 13 with $T(S_\lambda)_\mathbb{Q} \cong T$, the nine-dimensional representation of a Mumford fourfold, and $E = F$ a totally real cubic field, which may have Galois group S_3. By Proposition V.9(v) no composite of similarities reaches F, so Theorem V.3, the Fricke isogenies and every correspondence of the kinds listed are blind to it; by Proposition V.9(iv) a correspondence that does reach it has an irreducible component whose resolution has $p_g \geq 2$, transcendental rank at least 18 and $b_2 \geq 31$; and by Proposition V.10(iii) that component is not the graph of an element of a finite group acting on a cover of S_λ when F is not cyclic. The only correspondences of this size that we know to act on a transcendental space of K3 type are the dihedral ones, which give abelian fields, and the Kuga–Satake correspondence into the cohomology of an abelian variety of dimension 2^7, whose algebraicity is open [12, 41]; in Section VI we replace its abelian variety by a sixfold and show what the replacement does and does not give. Within quadratic fields the method now reaches $\sqrt{d}$ for $d \in \{2, 3, 5, 6, 7\}$ in families, for $d = 5, 7$ on the one-dimensional families of Theorem V.8; for $\mathbb{Q}(\sqrt{10})$, $\mathbb{Q}(\sqrt{14})$ and the other composite discriminants no family is known on which the chain of Theorem V.3 can be started.

VI. CONSTRUCTIONS AT (F2)

The first classes (F2) has to reach are the Weil classes of a very general abelian $2n$-fold of Weil type with $n \geq 3$, and the exceptional classes of a Mumford fourfold. We build an object at each.

A. The polarised semiregularity criterion

Let A be a very general member of the polarised family $\mathcal{D}_{n,n}$ of abelian $2n$-folds of Weil type for an imaginary quadratic field K, with polarisation class θ and Weil plane $W(A)$. Then $\mathrm{Hdg}^*(A) = \mathbb{Q}[\theta] \oplus W(A)$ and $W(A) \cdot \theta = 0$ [7, Thm. 4.14]. A class $\gamma \in H^{\mathrm{ev}}(A)$ has the polarised shape if

$$\gamma = N\omega + \sum_{k=0}^{2n} c_k \theta^k, \qquad N \neq 0,\ 0 \neq \omega \in W(A),\ c_k \in \mathbb{Q}.$$

Its Hankel rank $\rho \in \{0, 1, 2, 3\}$ is the rank of the Hankel matrix of the sequence $(k!\, c_k)_k$ capped at three; it is the number of distinct exponentials $e^{a\theta}$ needed to write the polynomial part when that number is at most two, and 3 otherwise.

Theorem VI.1 (The polarised criterion [7])**.** *Let $\mathcal{E}$ be a perfect complex on A with $\mathrm{ch}(\mathcal{E}) = \gamma$ of polarised shape and Hankel rank ρ. Then*

$$\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \ \geq\ r(\gamma) := (4 + \rho)\, n^2 - 2n,$$

and if equality holds the semiregularity map of $\mathcal{E}$ is injective, $\mathcal{E}$ deforms with A along the whole of $\mathcal{D}_{n,n}$, and the Weil classes are algebraic on every member of the family with full Hodge group. If some $c_k \neq 0$ with $1 \leq k \leq 2n-1$, the first-order Hodge locus of γ in $H^1(A, T_A)$ is the tangent space of $\mathcal{D}_{n,n}$, of dimension n^2. A perfect complex whose Chern character is exactly a multiple of a Weil class does not exist on a very general member, so the polynomial part is forced.

The number $r(\gamma)$ is the rank of the contraction $HT^2(A) \to H^*(A)$, $\xi \mapsto \xi \lrcorner \gamma$, on the Hochschild cohomology $HT^2 = \bigoplus_{p+q=2} H^q(A, \bigwedge^p T_A)$, of dimension $\binom{4n}{2}$; its kernel is the annihilator $\mathrm{Ann}(\gamma)$, which meets $H^1(A, T_A)$ in the first-order Hodge locus. The semiregularity map $\sigma_\mathcal{E}$ of a perfect complex composes with the characteristic map $\mathrm{ev}_\mathcal{E} : HT^2 \to \mathrm{Ext}^2(\mathcal{E}, \mathcal{E})$, $\xi \mapsto \xi \lrcorner \mathrm{At}(\mathcal{E})$, to the contraction into $\mathrm{ch}(\mathcal{E})$; this is how the lower bound arises, and it is the form in which the criterion is used below. Figure 5 shows $r(\gamma)$; the two values that matter below are $r = 57$ at $n = 3$, $\rho = 3$ and $r = 104$ at $n = 4$, $\rho = 3$. On the product E_0^6 of six copies of the elliptic curve $\mathbb{C}/\mathbb{Z}[i]$ every twisted complex of line bundles has $\dim \mathrm{Ext}^2 \geq 69 > 57$ [7, Thm. 18.35]; the criterion is not met there by any such object.

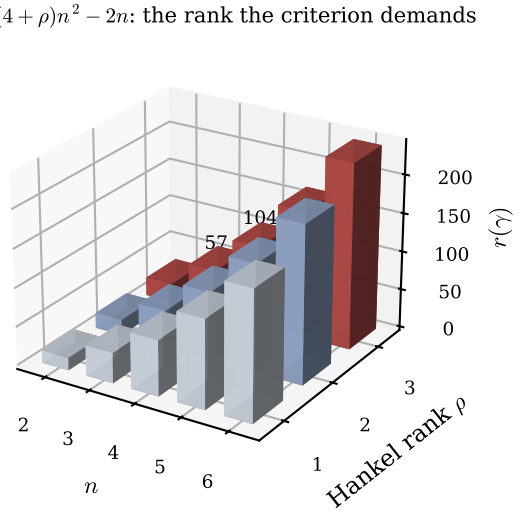


FIG. 5. The lower bound $r(\gamma) = (4 + \rho)n^2 - 2n$ of Theorem VI.1 for $2 \leq n \leq 6$ and Hankel rank $\rho = 1, 2, 3$. The two labelled bars are the product cases of Theorem VI.3.

B. The Weil plane of a product

Lemma VI.2. *Let A, A' be of Weil type for the same field K, of dimensions $2m$, $2m'$, with the diagonal K-action on $A \times A'$. Then $W(A \times A')$ is a two-dimensional*

rational subspace of the four-dimensional space $W(A)\otimes W(A')\subseteq H^{2n}(A\times A')$, $n=m+m'$, *and every rational* $w\in W(A\times A')$ *is* $\sum_{i,j}a_{ij}\,\omega_i\otimes\omega'_j$ *with* $a_{ij}\in\mathbb{Q}$ *for rational bases* (ω_i), (ω'_j) *of* $W(A)$, $W(A')$.

Proof. Over $\mathbb{C}$, $H^1(A\times A')=V\oplus V'$ splits into eigenspaces $V_\pm\oplus V'_\pm$ of K, and $\bigwedge_K^{2n}$ of the sum is spanned by $\det V_+\otimes\det V'_+$ and $\det V_-\otimes\det V'_-$; these lie in the span of the four products $\omega_\pm\otimes\omega'_\pm$ of the eigenvectors of $W(A)\otimes\mathbb{C}$ and $W(A')\otimes\mathbb{C}$. Both $W(A\times A')$ and $W(A)\otimes W(A')$ are defined over $\mathbb{Q}$, so the inclusion holds over $\mathbb{Q}$. The explicit formula is [7, Eq. (36)]: $\omega_1(A\times A')=\omega_1\otimes\omega'_1-d\,\omega_2\otimes\omega'_2$ for $K=\mathbb{Q}(\sqrt{-d})$. □

C. An object with the polarised Chern character

Theorem VI.3 (The product object)**.** *Let* A, A' *be very general in their polarised Weil families, with ample classes* θ_A, $\theta_{A'}$, $\theta=\theta_A+\theta_{A'}$, *and suppose that* $W(A)$ *and* $W(A')$ *are algebraic. Then there is a perfect complex* $\mathcal{E}$ *on* $A\times A'$, *a finite direct sum with shifts of exterior products*

$$F_i\boxtimes G'_j,\quad F_i\boxtimes L'^{\,k},\quad L^l\boxtimes G'_j,\quad L^l\boxtimes L'^{\,k},$$

with F_i, G'_j *coherent sheaves on* A, A' *and* L, L' *the ample bundles, such that*

$$\operatorname{ch}(\mathcal{E})=N\,w+\sum_{k=0}^{2n}c_k\theta^k,\qquad N\neq0,\ 0\neq w\in W(A\times A'),$$

has the polarised shape with Hankel rank $\rho=3$, *so that* $r(\operatorname{ch}\mathcal{E})=7n^2-2n$, *and the first-order Hodge locus of* $\operatorname{ch}(\mathcal{E})$ *in* $H^1(A\times A',T)$ *is the tangent space of* $\mathcal{D}_{n,n}$.

The hypotheses hold in the two open cases nearest to the known ones: A, A' very general Weil fourfolds of trivial discriminant, where Markman proves the Weil classes algebraic [25, 26], so that $A\times A'$ is an eightfold with $n=4$; and A such a fourfold and $A'=B$ a Weil surface of any discriminant d, whose Weil plane is of type $(1,1)$ and algebraic by Lefschetz, so that $A\times B$ is a sixfold of discriminant d with $n=3$.

Proof. Since $\mathrm{Hdg}^*(A)=\mathbb{Q}[\theta_A]\oplus W(A)$, algebraic classes on A are rational combinations of Chern characters of coherent sheaves, and the Chern character of a coherent sheaf is a Hodge class, the algebraicity of $W(A)$ gives coherent sheaves F_1, F_2 with $\operatorname{ch}(F_i)=N_i\omega_i+p_i(\theta_A)$, $N_i\neq0$, for a basis (ω_1,ω_2) of $W(A)$; likewise G'_1, G'_2 on A' with $\operatorname{ch}(G'_j)=N'_j\omega'_j+p'_j(\theta_{A'})$. Fix $0\neq w\in W(A\times A')$ rational, $w=\sum a_{ij}\omega_i\otimes\omega'_j$ by Lemma VI.2, and a common denominator M of the numbers $Ma_{ij}/(N_iN'_j)$, so that these are integers. Negative multiplicities are realised by the shift [1], which changes the sign of ch.

Step 1. $\mathcal{E}_0=\bigoplus_{i,j}(F_i\boxtimes G'_j)^{\oplus Ma_{ij}/(N_iN'_j)}$ has, by multiplicativity,

$$\operatorname{ch}(\mathcal{E}_0)=Mw+\sum_i\omega_i\otimes q'_i(\theta_{A'})+\sum_j q_j(\theta_A)\otimes\omega'_j+s,$$

with q'_i, q_j polynomials and $s\in\mathbb{Q}[\theta_A]\otimes\mathbb{Q}[\theta_{A'}]$.

Step 2. The classes $e^{k\theta_{A'}}=\operatorname{ch}(L'^{\,k})$, $k=0,\dots,2m'$, form a basis of $\mathbb{Q}[\theta_{A'}]/(\theta_{A'}^{2m'+1})$: the matrix $(k^j/j!)_{j,k}$ has determinant $\prod_j(1/j!)$ times the Vandermonde determinant of $0,1,\dots,2m'$, which is nonzero. Write $q'_i=\sum_k b_{ik}e^{k\theta_{A'}}$ and add $(F_i\boxtimes L'^{\,k})^{\oplus -Mb_{ik}/N_i}$, after enlarging M to clear denominators; this removes $\omega_i\otimes q'_i$ and adds only terms $p_i(\theta_A)\otimes e^{k\theta_{A'}}$, which lie in $\mathbb{Q}[\theta_A]\otimes\mathbb{Q}[\theta_{A'}]$. Symmetrically remove $q_j\otimes\omega'_j$ with summands $L^l\boxtimes G'_j$.

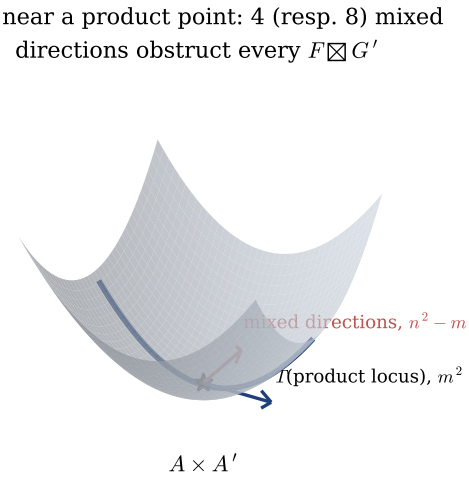


FIG. 6. The domain $\mathcal{D}_{n,n}$ near a product point $A\times A'$. The curve is the product locus, whose tangent space has dimension $m^2+m'^2$; the $2mm'$ mixed directions obstruct every exterior product $F\boxtimes G'$, and only the combined object of Theorem VI.3 has the full tangent space as first-order Hodge locus.

Step 3. The classes $e^{l\theta_A+k\theta_{A'}}=\operatorname{ch}(L^l\boxtimes L'^{\,k})$ form a basis of $\mathbb{Q}[\theta_A]/(\theta_A^{2m+1})\otimes\mathbb{Q}[\theta_{A'}]/(\theta_{A'}^{2m'+1})$ by the same determinant, so the remaining part in $\mathbb{Q}[\theta_A]\otimes\mathbb{Q}[\theta_{A'}]$ is cancelled by line bundles with integer multiplicities and shifts. Finally add $(L\boxtimes L')^{\otimes k}$, $k=0,1,2$, with generic nonzero multiplicities μ_k. The result is

$$\operatorname{ch}(\mathcal{E})=Mw+\sum_{k=0}^{2}\mu_k e^{k\theta},$$

a polarised shape whose polynomial part is a sum of three distinct exponentials with nonzero coefficients. Three exponentials are not one of the shapes of Hankel rank at most two, so $\rho=3$; and some c_k with $1\le k\le 2n-1$ is nonzero, so the first-order Hodge locus is the tangent space of $\mathcal{D}_{n,n}$ by Theorem VI.1. □

The construction was carried out in the exterior-algebra model of $H^*(A\times A')$ for $(m,m')=(2,1)$ and $(2,2)$: the cross terms of $\operatorname{ch}(\mathcal{E}_0)$ vanish, nothing is left outside $W(A\times A')\oplus\mathbb{Q}[\theta]$ after Steps 2 and 3, the Hankel rank is 3, and the first-order Hodge locus has dimension 9, resp. 16, equal to n^2 (Appendix A).

D. What a direct sum cannot do

The tangent space of $\mathcal{D}_{n,n}$ at a product point contains the tangent space of the product locus $\mathcal{D}_{m,m}\times\mathcal{D}_{m',m'}$, of dimension $m^2+m'^2$, and $n^2-m^2-m'^2=2mm'$ mixed directions. In the same model, each exterior product $F\boxtimes G'$ of sheaves of polarised shape has first-order Hodge locus inside $T\mathcal{D}_{n,n}$ equal to the tangent space of the product locus: all $2mm'$ mixed directions obstruct it, four for $A\times B$ and eight for $A\times A'$ (Fig. 6). The object $\mathcal{E}$ of Theorem VI.3 has the whole tangent space as its first-order locus because its Chern character is a single polarised class and not a product. Semiregularity is a different matter.

Proposition VI.4 (Direct sums are never semiregular)**.**

(i) *Let $\mathcal{E}=\bigoplus_k\mathcal{E}_k$ with at least two nonzero summands. Then the semiregularity map $\sigma_{\mathcal{E}}$ vanishes on $\bigoplus_{k\neq l}\operatorname{Ext}^2(\mathcal{E}_k,\mathcal{E}_l)$. So $\sigma_{\mathcal{E}}$ is injective only if $\operatorname{Ext}^2(\mathcal{E}_k,\mathcal{E}_l)=0$ for all $k\neq l$.*

(ii) *For every direct-sum realisation $\mathcal{E}$ of Theorem VI.3, $\sigma_{\mathcal{E}}$ is not injective and $\dim\operatorname{Ext}^2(\mathcal{E},\mathcal{E})>r(\mathrm{ch}\mathcal{E})$.*

Proof. (i) The semiregularity map of Buchweitz and Flenner is built from the traces $\xi\mapsto\operatorname{tr}\big(\xi\circ\mathrm{At}(\mathcal{E})^p\big)$ of compositions with powers of the Atiyah class [5, Sect. 4]. The Atiyah class of a direct sum is block diagonal, so for ξ in an off-diagonal block $\operatorname{Ext}^2(\mathcal{E}_k,\mathcal{E}_l)$ the composition $\xi\circ\mathrm{At}(\mathcal{E})^p$ is off-diagonal, and the trace of an off-diagonal block is zero.

(ii) The summands of $\mathcal{E}$ include $F_1\boxtimes L'^{\,k}$ and $F_1\boxtimes L'^{\,l}$ for two exponents $k<l$, because Step 2 of the proof uses every power $L'^{\,k}$ with $0\le k\le 2m'$ (the exponentials form a basis and q_1' is a nonzero polynomial of positive degree for a sheaf of polarised shape, which has p_1 of positive degree by Theorem VI.1). By the Künneth formula

$$\operatorname{Ext}^2(F_1\boxtimes L'^{\,k},F_1\boxtimes L'^{\,l})\supseteq\operatorname{Ext}^2(F_1,F_1)\otimes H^0(A',L'^{\,l-k}),$$

and $\dim\operatorname{Ext}^2(F_1,F_1)\ge r(\mathrm{ch}F_1)\ge 12$ by Theorem VI.1 on the fourfold A ($n=2$, $\rho\ge 0$), while H^0 of an ample bundle is nonzero. So an off-diagonal block is nonzero, $\sigma_{\mathcal{E}}$ has a kernel by (i), and $\dim\operatorname{Ext}^2(\mathcal{E},\mathcal{E})>r$ because equality would force injectivity. □

Proposition VI.5 (Convolutions)**.** *Let $\mathcal{E}_\delta$ be a one-sided twisted complex on pieces $P_1,\dots,P_N$ with shifts s_a, ordered so that the differential $\delta=(\delta_{ab})_{a<b}$, $\delta_{ab}\in\operatorname{Ext}^{1+s_a-s_b}(P_a,P_b)$, is strictly upper triangular, and let $\gamma=\mathrm{ch}(\mathcal{E}_\delta)=\sum_a(-1)^{s_a}\mathrm{ch}(P_a)$.*

(i) *The semiregularity map $\sigma_{\mathcal{E}_\delta}$ vanishes on every class of $\operatorname{Ext}^2(\mathcal{E}_\delta,\mathcal{E}_\delta)$ represented by a strictly upper triangular cocycle. So the criterion holds for $\mathcal{E}_\delta$ only if every strictly upper triangular cocycle of degree two is a coboundary.*

(ii) *Suppose no term of the twisted differential carries a class of degree one into a diagonal block $\operatorname{Ext}^*(P_c,P_c)$. Let $v\in T\mathcal{D}_{n,n}$ have a nonzero mixed component. If some piece $P=X\boxtimes Y$ has $\operatorname{rk}Y\,c_1(X)/\theta_A\neq\operatorname{rk}X\,c_1(Y)/\theta_{A'}$, then $\mathrm{ev}_{\mathcal{E}_\delta}(v)\neq 0$ and $\sigma_{\mathcal{E}_\delta}(\mathrm{ev}_{\mathcal{E}_\delta}(v))=0$: the criterion fails.*

(iii) *Suppose the pieces split into two nonempty groups with $\operatorname{Ext}^*(P_a,P_b)=0$ in both directions for a, b in different groups. Then $\mathcal{E}_\delta=\mathcal{E}'\oplus\mathcal{E}''$ is the direct sum of the convolutions of the groups, and the criterion holds for $\mathcal{E}_\delta$ only if every $v\in T\mathcal{D}_{n,n}$ annihilates $\mathrm{ch}(\mathcal{E}')$ and $\mathrm{ch}(\mathcal{E}'')$ separately.*

Proof. (i) The components of $\sigma_{\mathcal{E}_\delta}(\xi)$ are the traces $\operatorname{tr}(\xi\circ\mathrm{At}(\mathcal{E}_\delta)^p)/p!$ [5, Sect. 4]. In the harmonic model of a twisted complex the Atiyah class is represented by $[\nabla,\bar\partial+\delta]$ with ∇ the direct sum of the Chern connections of the pieces; it is block upper triangular with diagonal blocks $\mathrm{At}(P_a)$ and off-diagonal blocks $\nabla\delta_{ab}$. The homotopy transfer $\xi\mapsto\sum_k(-h[\delta,\cdot])^k\xi$ that produces the chain-level representative of ξ raises or preserves the block position $b-a$, so a strictly upper triangular cocycle has a strictly upper triangular representative. The product of an upper triangular block matrix with a strictly upper triangular one is strictly upper triangular, and its trace, the sum of the traces of its diagonal blocks, is zero. The trace is a chain map for $\bar\partial+[\delta,\cdot]$ by cyclicity, so the vanishing holds in cohomology. Since $\sigma_{\mathcal{E}_\delta}\circ\mathrm{ev}_{\mathcal{E}_\delta}$ is contraction into γ, of rank $r(\gamma)$, the kernel of $\sigma_{\mathcal{E}_\delta}$ contains the image of the strictly upper triangular classes and $\dim\operatorname{Ext}^2\ge r(\gamma)+\dim(\text{that image})$.

(ii) The Atiyah class of $X\boxtimes Y$ is $\mathrm{At}(X)\otimes 1+1\otimes\mathrm{At}(Y)$, so the diagonal component of $\mathrm{ev}_{\mathcal{E}_\delta}(v)$ at P is $v\lrcorner\mathrm{At}(P)$, with trace $v\lrcorner\mathrm{ch}_1(P)$. Write $v=v_A+v_{A'}+z\otimes d'+z'\otimes d$ with $v_A\in T\mathcal{D}_{m,m}$, $v_{A'}\in T\mathcal{D}_{m',m'}$, $z\in H^{0,1}(A)$, $d'\in H^0(T_{A'})$, $z'\in H^{0,1}(A')$, $d\in H^0(T_A)$. Since v kills $\theta=\theta_A+\theta_{A'}$ and $v_A\lrcorner\theta_A=v_{A'}\lrcorner\theta_{A'}=0$, we have $v\lrcorner\theta_{A'}=-v\lrcorner\theta_A=-z'\wedge(d\lrcorner\theta_A)$, nonzero when $d\neq 0$ because θ_A is nondegenerate and $z'\wedge$ is injective on $H^{0,1}(A)$; the case $d=0$, $d'\neq 0$ is symmetric. On a very general member $c_1(X)$ and $c_1(Y)$ are multiples of the polarisations, so $v\lrcorner\mathrm{ch}_1(P)=\big(\operatorname{rk}Y\,c_1(X)/\theta_A-\operatorname{rk}X\,c_1(Y)/\theta_{A'}\big)\,v\lrcorner\theta_A\neq 0$ under the hypothesis on P. Under the assumption on the differential, the diagonal components of a class in $\operatorname{Ext}^2(\mathcal{E}_\delta,\mathcal{E}_\delta)$ are well defined, so $\mathrm{ev}_{\mathcal{E}_\delta}(v)\neq 0$, while $\sigma_{\mathcal{E}_\delta}(\mathrm{ev}_{\mathcal{E}_\delta}(v))=v\lrcorner\gamma=0$ because v is tangent to the Hodge locus of γ.

(iii) With no Ext groups between the groups there are no components of δ between them, so the twisted complex is block diagonal and $\operatorname{Ext}^2(\mathcal{E}_\delta,\mathcal{E}_\delta)=\operatorname{Ext}^2(\mathcal{E}',\mathcal{E}')\oplus\operatorname{Ext}^2(\mathcal{E}'',\mathcal{E}'')$ with $\sigma_{\mathcal{E}_\delta}=\sigma_{\mathcal{E}'}\oplus\sigma_{\mathcal{E}''}$. For each summand the contraction into its Chern character factors through ev and σ, so that $\dim\operatorname{Ext}^2(\mathcal{E}',\mathcal{E}')\ge\operatorname{rank}(\xi\mapsto\xi\lrcorner\mathrm{ch}\mathcal{E}')$, and therefore $\dim\operatorname{Ext}^2(\mathcal{E}_\delta,\mathcal{E}_\delta)\ge\dim HT^2-\dim\big(\mathrm{Ann}(\mathrm{ch}\mathcal{E}')\cap\mathrm{Ann}(\mathrm{ch}\mathcal{E}'')\big)$, and $r(\gamma)=\dim HT^2-\dim\mathrm{Ann}(\gamma)$ with $\mathrm{Ann}(\gamma)=T\mathcal{D}_{n,n}$. Equality $\dim\operatorname{Ext}^2=r(\gamma)$ forces the common annihilator to be all of $T\mathcal{D}_{n,n}$. □

For the pieces of Theorem VI.3 the relevant splitting in (iii) is between the pieces $F_i \boxtimes Y$ carrying a secant sheaf on A and the pieces $L^l \boxtimes Y'$, which is the case when $\mathrm{Ext}^*(F_i, L^l) = \mathrm{Ext}^*(L^l, F_i) = 0$ for the powers used; then $\mathrm{ch}(\mathcal{E}') = MN'w + \sum_j q_j \otimes \omega'_j$ and $\mathrm{ch}(\mathcal{E}'') = \gamma - \mathrm{ch}(\mathcal{E}')$, and the criterion needs every mixed direction of $T\mathcal{D}_{n,n}$ to annihilate $\sum_j q_j \otimes \omega'_j$. In the exterior-algebra model the common annihilator is the tangent space of the product locus, of dimension $m^2 + m'^2$, so that $\dim \mathrm{Ext}^2 \geq r + 2mm'$ and the convolution fails; we have not written a hand proof of that rank, and nothing below uses it. Part (ii) applies in particular to every piece $L^l \boxtimes L'^k$ with $l \neq k$, which Step 3 of Theorem VI.3 uses.

E. Where the product object stops

Every test that depends on $\mathrm{ch}(\mathcal{E})$ alone is passed: polarised shape, $N \neq 0$, Hankel rank 3, first-order Hodge locus equal to $T\mathcal{D}_{n,n}$, lower bound 57 or 104. The criterion then asks for $\dim \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) = r$, Proposition VI.4 shows that no direct sum achieves it, and Proposition VI.5 says what a convolution of the same pieces must do: kill every strictly upper triangular class of degree two, carry classes of degree one from the lower triangle into the diagonal blocks of every piece with mismatched slopes, and, if the secant pieces and the line-bundle pieces on a factor are orthogonal, make the mixed directions annihilate $\sum_j q_j \otimes \omega'_j$. All three demands are statements about the groups $\mathrm{Ext}^*(F_1, F_2)$, $\mathrm{Ext}^*(F_i, L^l)$, $\mathrm{Ext}^*(L^l, F_i)$ and their analogues on A'. Those groups are not computed in [25–27], where the only object with known Ext groups on a very general Weil fourfold of trivial discriminant is the twisted transform $\Phi(F \boxtimes F^\vee)$ with $\dim \mathrm{Ext}^k = 1, 8, 18, 8, 1$, whose Brauer class is half the Poincaré class and which therefore cannot be mixed with untwisted line bundles; and we have not computed them. This is the exact stopping point: the polarised criterion at a product point is a finite question about the Ext algebra of the secant sheaves, and Proposition VI.5 lists what the answer must contain for the route to stay open. The analogous question on E_0^6, for line bundles, closed negatively with $\dim \mathrm{Ext}^2 \geq 69$ [7, Thm. 18.35]; that is an analogy and not a proof of the answer here.

F. The exceptional classes of a Mumford fourfold

Keep Setting III.5 and write ψ for the polarisation form on V, $T = D^0$, $q_\lambda(x, y) = \mathrm{Tr}_{F/\mathbb{Q}}(\lambda\, \mathrm{trd}(xy))$ for $\lambda \in F$, and m_a for multiplication by $a \in F$ on T. The copy $T_X(-1) \subseteq V \otimes V$, $x \mapsto \big((v, w) \mapsto \psi(xv, w)\big)$, is symmetric and carries the form induced by $\psi \otimes \psi$, equal to $-4\,\mathrm{trd}(xy)$ summed over the three real places, that is q_1 up to the scalar -4. The exceptional classes of $X \times X$ are the elements m_a, $a \in F \setminus \mathbb{Q}$, of $\mathrm{End}_{\mathrm{Hdg}} T_X(-1) = F$ [7, Prop. 21.2], and in [7, Thm. 21.4] their algebraicity is derived from the Kuga–Satake class of a K3 surface S_λ with $T(S_\lambda)_\mathbb{Q} \cong (T, q_\lambda)$. We replace S_λ by an abelian variety.

Setting VI.6 (The sixfold Z_K)**.** Let $K = F(\sqrt{-\delta}) \subseteq D$ be a CM quadratic extension of F inside D, $\delta \in F$ totally positive; every maximal commutative CM subfield of D is of this form since D is definite at two real places. Let D carry the left action of $G = D^1$ and the right action of the torus $K^1 = \ker(N_{K/F})$. Over $\bar{\mathbb{Q}}$, $D = D_1 \oplus D_2 \oplus D_3$ with $D_i = V_i \otimes W_i$, V_i the standard representation of the i-th factor of G and $W_i = \chi_i \oplus \chi_i^{-1}$ that of the i-th factor of K^1. Fix a point t of the Shimura curve C of X, with Hodge cocharacter μ_X nontrivial only at the split place σ_3, and let $\mu_Z = \mu'\mu''$, with μ' the action of μ_X by left multiplication and μ'' the cocharacter of K^1 of weight $\pm\frac{1}{2}$ on $\chi_1^{\pm1}$, $\chi_2^{\pm1}$ and 0 on $\chi_3^{\pm1}$. Then $D_\mathbb{C}$ has type $(1,0)$ on $V_1\chi_1$, $V_2\chi_2$, $V_3^{1,0}W_3$ and $(0,1)$ on the complements, so D carries a rational Hodge structure of weight one with $h^{1,0} = 6$ and Hodge group inside $G \cdot K^1$. For $c \in K$ with $\bar{c} = -c$ put $\psi_c(x, y) = \mathrm{Tr}_{F/\mathbb{Q}}\mathrm{trd}(\bar{x}\, y\, c)$. The abelian sixfold $Z_K = Z_{K,t}$ is the one with $H^1(Z_K) = D$ and polarisation ψ_c for c in the cone where ψ_c is positive; over a finite cover of C these form an abelian scheme [37], and we use only the fibre at t.

Lemma VI.7. *ψ_c is alternating, invariant under left multiplication by D^1 and right multiplication by K^1, and the ψ_c, $c \in K$ with $\bar{c} = -c$, fill the three-dimensional space of all $G \times K^1$-invariant alternating forms on D. Hence $\mathrm{NS}(Z_K)_\mathbb{Q}$ has dimension three.*

Proof. The reduced trace is invariant under conjugation and cyclic, so $\psi_c(y, x) = \mathrm{Trtrd}(\bar{y}\, x\, c) = \mathrm{Trtrd}(\bar{c}\, \bar{x}\, y) = \mathrm{Trtrd}(\bar{x}\, y\, \bar{c})$, and ψ_c is alternating exactly when $\bar{c} = -c$. For g with $\mathrm{nrd}(g) = 1$, $\mathrm{trd}(\overline{gx}\, gy\, c) = \mathrm{trd}(\bar{x}\, \bar{g} g\, y\, c) = \mathrm{trd}(\bar{x}\, y\, c)$. For $k \in K^1$, $\mathrm{trd}(\overline{xk}\, yk\, c) = \mathrm{trd}(\bar{k}\, \bar{x}\, y\, k\, c) = \mathrm{trd}(\bar{x}\, y\, k\, c\, \bar{k}) = \mathrm{trd}(\bar{x}\, y\, c)$ because K is commutative and $k\bar{k} = 1$. Over $\bar{\mathbb{Q}}$ the invariant alternating forms live in the three summands $\bigwedge^2 (V_i \otimes W_i)^{G \times K^1} = \bigwedge^2 V_i \otimes (\chi_i \chi_i^{-1})$, one dimension each, and the cross terms $D_i \otimes D_j$ carry the nontrivial characters $\chi_i^{\pm1}\chi_j^{\pm1}$; so the space has dimension $3 = \dim_\mathbb{Q} K^-$, which the ψ_c fill since $c \mapsto \psi_c$ is injective. Hodge classes of degree two are the invariant alternating forms, as the Hodge group lies in $G \cdot K^1$ and the invariants of $G \cdot K^1$ are the three forms, which are of type $(1,1)$. □

Lemma VI.8 (Where T sits in Z_K)**.**

(i) *The K^1-invariants of $\bigwedge^2 D_i$ are four-dimensional: $S^2 V_i \otimes \bigwedge^2 W_i$, the adjoint representation of $\mathfrak{sl}(V_i)$, and $\bigwedge^2 V_i \otimes \mathbf{1}$. The sum T_Z of the three adjoint pieces is defined over $\mathbb{Q}$, is a sub-Hodge structure of $H^2(Z_K)$, and is isomorphic, as a Hodge structure with G-action, to $T_X(-1)$.*

(ii) *The form induced on $\bigwedge^2 D$ by ψ_c restricts on T_Z to q_{c^2}, up to a rational scalar independent of the place; here $c^2 = -N_{K/F}(c) = -f^2\delta \in F$ for $c = f\sqrt{-\delta}$.*

Proof. (i) $\bigwedge^2(V_i\otimes W_i)=S^2V_i\otimes\bigwedge^2W_i\oplus\bigwedge^2V_i\otimes S^2W_i$; the torus acts trivially on $\bigwedge^2W_i=\chi_i\chi_i^{-1}$ and on one summand of $S^2W_i=\chi_i^2\oplus\mathbf{1}\oplus\chi_i^{-2}$. The cross terms carry nontrivial characters. The Galois group permutes the places, so T_Z is defined over $\mathbb{Q}$; as a G-representation it is the adjoint representation T. The cocharacter μ'' acts trivially on K^1-invariants and the centre acts on $\bigwedge^2D$ with weight two, so the Hodge structure on T_Z is the one induced by the adjoint action of μ', twisted to weight two; this is the recipe that gives $T_X(-1)$ at the same point t.

(ii) By Schur's lemma the induced form is, on each irreducible T_i, a multiple of $\mathrm{trd}(xy)$. In the split model at the place i the element c acts on D_i by $\sigma_i(c)\,\mathrm{diag}(1,-1)$, whose square is $\sigma_i(c^2)$; the multiple is $\sigma_i(c)^2$ times a rational constant that is the same at every place, since the computation in the weight basis is identical at the three places. The sum over the places is q_{c^2} up to that constant. □

The sixfold link over the Shimura curve

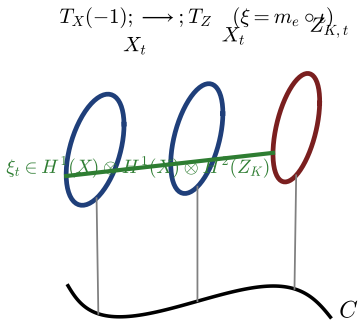


FIG. 7. The sixfold link. Two copies of the Mumford fourfold X_t and the sixfold $Z_{K,t}$ sit over the same point t of the Shimura curve C; the class ξ_t in the mixed Künneth component identifies the two copies of T and is flat along C.

Theorem VI.9 (The sixfold link)**.** *Let ξ be a nonzero element of* $\mathrm{Hom}_{\mathrm{Hdg}}(T_X(-1),T_Z)\subseteq H^4(X\times X\times Z_K)$.

(i) $\mathrm{Hom}_{\mathrm{Hdg}}(T_X(-1),T_Z)=F\cdot\iota$ *for the identification ι of Lemma VI.8, so $\xi=m_e\circ\iota$ for a unique $e\in F^\times$. The $G\times K^1$-invariant classes of degree four on $X\times X\times Z_K$ have dimension* 26*, and ξ and $\psi\otimes\mathrm{NS}(Z_K)$ are the only ones in the Künneth component $H^1(X)\otimes H^1(X)\otimes H^2(Z_K)$.*

(ii) *If ξ is algebraic, then $\xi^\dagger\xi$, the composite with the adjoint for the Lefschetz pairing on $X\times X$ and the intersection pairing on Z_K, is an algebraic endomorphism of $T_X(-1)$ equal to $m_{a(e)}$ with $a(e)=\kappa\,e^2c^2\in F$, $\kappa\in\mathbb{Q}^\times$.*

(iii) *If $a(e)\notin\mathbb{Q}$, the exceptional classes of $X\times X$ are algebraic at t, hence at every point of C, and so is every Hodge class on every power of X.*

(iv) *$a(e)\notin\mathbb{Q}$ for every $e\in F^\times$ if and only if K contains no imaginary quadratic field. If $K=F\cdot k$ with $k=\mathbb{Q}(\sqrt{-d})$ and $c=f\sqrt{-d}$, then $a(e)\in\mathbb{Q}$ exactly for $e\in\mathbb{Q}f^{-1}$.*

(v) *Conversely, if the exceptional classes are algebraic and some algebraic cycle on $X^r\times Z_K$ acts nonzero on T_Z, then every ξ is algebraic.*

Proof. (i) Both sides are the irreducible G-representation T with $\mathrm{End}_{\mathrm{Hdg}}=F$ [7, Prop. 21.2], so $\mathrm{Hom}_{\mathrm{Hdg}}=F\iota$. The count is the dimension of the kernel of the six raising and lowering operators of $\mathfrak{sl}_2^3$ on the torus-weight-zero part of $\bigwedge^4(V\oplus V\oplus D)$, computed by Künneth type: $(1,1,2)$ gives $3+3$, the line $F\iota$ and $\psi\otimes\mathrm{NS}(Z_K)$; $(2,2,0)$ gives $4=\mathbb{Q}\oplus F$, the Hodge classes of $X\times X$ in $H^2\otimes H^2$; $(0,0,4)$ gives $6=S^2\mathrm{NS}(Z_K)$; $(2,0,2)$ and $(0,2,2)$ give 3 each; and $(4,0,0)$, $(0,4,0)$, $(3,1,0)$, $(1,3,0)$ one each, products of divisor classes. The sum is 26. The Hodge group of the product lies in $G\cdot K^1$, so 26 bounds the Hodge classes from above; equality is not needed.

(ii) The adjoint of a correspondence for the Lefschetz pairing on $H^2(X\times X)$ and the polarised intersection pairing on Z_K is the transposed cycle composed with cup products with powers of the polarisations and the inverse Lefschetz isomorphism, which is algebraic on an abelian variety [20, §2]; so $\xi^\dagger\xi$ is algebraic. By Lemma VI.8, $T_X(-1)$ carries $-4q_1$ and T_Z carries a rational multiple of q_{c^2}, and for $\xi=m_e$, $q_{c^2}(m_ex,m_ey)=\mathrm{Tr}(c^2e^2\mathrm{trd}(xy))=q_1(c^2e^2x,y)$, so $\xi^\dagger\xi=\kappa\,m_{e^2c^2}$.

(iii) An irrational $a\in F$ generates the cubic field, so every m_b, $b\in F$, is a polynomial in $\xi^\dagger\xi$, algebraic on $T_X(-1)$. Compose with the product map $\mu\colon H^2(X\times X)\supseteq T_X(-1)\to H^2(X)$ and its adjoint, both built from ψ, the diagonal and θ and hence algebraic [7, Lem. 21.3]: the composite $\mu\circ(m_b\otimes 1)\circ\mu^\dagger$ acts on the primitive part U of $H^2(X)$ by the element $\nu(\mathrm{Tr}b-b)/2$ of F, irrational when b is. With the identity and the projector onto $\mathbb{Q}\theta$ this generates $\mathbb{Q}\times F=\mathrm{End}_{\mathrm{Hdg}}H^2(X)$, so by [7, Prop. 21.2] the exceptional classes are algebraic. The class ξ is flat along C and so are the exceptional classes, so by Lemma IV.2 algebraicity at one point of the curve C gives it on a countable set or everywhere; and [7, Cor. 20.30] shows that algebraicity at one point of C gives it at every point, and then every Hodge class on every power of X is algebraic [7, Thm. 21.7].

(iv) $a(e)=\kappa e^2f^2\delta$ is rational iff $e^2f^2\delta\in\mathbb{Q}$. If $-\delta\notin\mathbb{Q}^\times(F^\times)^2$ then $\delta\notin\mathbb{Q}^\times(F^\times)^2$ and $e^2f^2\delta\in\mathbb{Q}$ is impossible. If $-\delta=-dg^2$ with $d\in\mathbb{Q}^\times$, $g\in F^\times$, then with $c=f\sqrt{-d}$ one gets $a(e)=\kappa e^2f^2d$, rational iff $(ef)^2\in\mathbb{Q}$ iff $ef\in\mathbb{Q}$, a cubic field having no irrational element

with rational square. And $-\delta \in \mathbb{Q}^\times (F^\times)^2$ exactly when $K \supseteq \mathbb{Q}(\sqrt{-d})$ for some d.

(v) By [7, Thm. 21.7] the Hodge conjecture holds for all powers of X once the exceptional classes are algebraic. A cycle on $X^r \times Z_K$ acting nonzero on the irreducible T_Z carries it isomorphically onto a summand of $H^{2k}(X^r)(k-1)$; projecting onto that summand and composing with an isomorphism to $T_X(-1) \subseteq H^2(X \times X)$, a Hodge class on X^{r+2} and algebraic, gives an algebraic nonzero element of $\mathrm{Hom}_{\mathrm{Hdg}}(T_Z, T_X(-1)) = F$; composing with the algebraic m_a gives every element of that line, and transposing gives every ξ. □

Proposition VI.10 (Hyper-Kähler carriers give ± 1)**.** *Let $Y_1, \dots, Y_m$ be projective hyper-Kähler varieties of $K3^{[n_j]}$ type whose transcendental spaces are Hodge isometric to $(T, r_j q_{\lambda_j})$, $r_j \in \mathbb{Q}^\times$, $\lambda_j \in F$ totally positive. Every composite of rational Hodge isometries between them that returns to Y_j acts on T by ± 1, and every composite of Hodge similarities by an element of $\mathbb{Q}$.*

Proof. A Hodge isometry $(T, r_j q_{\lambda_j}) \to (T, r_k q_{\lambda_k})$ is a Hodge endomorphism of T, hence m_a for some $a \in F$, and it is an isometry iff $r_j \lambda_j = r_k \lambda_k a^2$. A composite returning to j is m_a with $a^2 = 1$, and a composite of similarities is m_a with $a^2 \in \mathbb{Q}$, which in a cubic field forces $a \in \mathbb{Q}$. The isometries are algebraic by [8, 24], which is why the statement matters: they realise m_b between S_λ and $S_{\lambda b^{-2}}$ and between their moduli spaces of sheaves, but every return is trivial. □

G. Where the link stops

Theorem VI.9 replaces the Kuga–Satake variety of [7, Thm. 21.4], of dimension 2^7, by an abelian sixfold over the same curve, and it replaces the irrationality that had to be extracted from anticommuting Clifford blocks by the arithmetic of K: for every K that is not a compositum $F \cdot k$ every link closes the target. Its logical status is the same as the Kuga–Satake link: by (v) it is equivalent to the target together with one algebraic link, not a reduction to something smaller.

The missing step is the algebraicity of one class $\xi \in H^4(X \times X \times Z_K)$. It is exceptional: its component in $V \otimes V$ is the nine-dimensional T, not a multiple of ψ, so it is not invariant under $\mathrm{Sp}(V)$ and not in the Lefschetz algebra of $X \times X \times Z_K$ (Lemma III.6); the theorems that prove the conjecture when the Hodge ring is generated by divisors do not apply, and no Weil class of Z_K or of $X \times X$ is involved. At the points of complex multiplication of C the class ξ_c is a Hodge class on a CM abelian variety in a mixed Künneth component between $X_c \times X_c$ and $Z_{K,c}$ with $\mathrm{Hom}(X_c, Z_{K,c}) = 0$, so it is not a pull-back from one factor nor a product of lower-degree Hodge classes, and the generation theorems for CM abelian varieties do not reach it either. Unlike the Kuga–Satake class, the link is therefore not known even at the countably many special points. This is the price of staying inside abelian varieties, and it is the exact statement that (F2) needs at its smallest instance.

Hodge classes of a very general Weil $2n$-fold, $n = 3$: η^n and the Weil line carry different characters

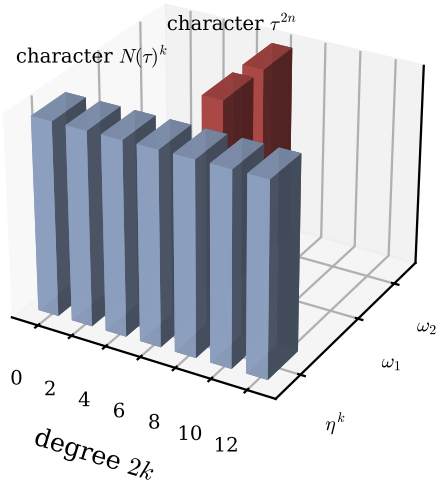


FIG. 8. The Hodge classes of a very general abelian sixfold of Weil type and the characters of $K^\times$ on them: η^k carries $N(\tau)^k$, the Weil plane carries τ^{2n}. An eigenclass for N^n in degree $2n$ is a multiple of η^n.

VII. WHERE THE EARLIER ATTEMPT FAILS

The preprint [3] of the two authors claimed the conjecture in full. We record here each point at which it fails, so that the present paper can be read as its replacement and not as a continuation. Numbers refer to that preprint.

a. The secant cycle (Definition 3, Lemma 4). For an abelian $2n$-fold A of Weil type the preprint takes $\Sigma = \mathrm{Sec}^n(A^\vee)$, the closure of the union of the $(n-1)$-planes spanned by n points of $A^\vee$ in a projective embedding, and the cycle $Z_\Sigma = \Sigma \cap A^\vee$, asserted to have codimension n. Every point of $A^\vee$ lies on such a plane, so $A^\vee \subseteq \Sigma$ and $Z_\Sigma = A^\vee$: the intersection is not proper. The sheaf $\mathcal{O}_\Sigma|_{A^\vee}$ is $\mathcal{O}_{A^\vee}$, whose Fourier–Mukai transform is the skyscraper sheaf at the origin shifted by $-2n$, with Chern character the class of a point. So $\mathrm{ch}_k = 0$ for $1 \le k < 2n$, contrary to Lemma 4, and Proposition 2 has nothing to identify with a Weil class.

b. The polarisation (Definition 3). The polarisation is taken principal, so $\chi(L) = 1$ and the embedding is into $\mathbb{P}^0$; the dimension count needs $\chi(L) = 2n^2 + 2n$. A principal polarisation is never very ample.

c. The dimension of the Weil space (Proposition 1). $\det_K H^1(A)$ is a two-dimensional $\mathbb{Q}$-space of Hodge classes [45], so at a very general point $\mathrm{Hdg}^n(A) = \mathbb{Q}\eta^n \oplus W(A)$ has dimension three, not two.

d. The eigencharacter (Lemmas 1, 2, Proposition 2, Theorem 4). Equation (1) of the preprint gives $\tau^*\eta = N(\tau)\eta$ for $\tau \in K^\times$, so η^n has character $N(\tau)^n$, while Proposition 2 asserts character 1. The group $K^\times$ acts on $W(A) = \bigwedge_K^{2n} H^1(A)$ by $\tau \mapsto \tau^{2n}$, which differs from $N(\tau)^n = (\tau\bar\tau)^n$ unless $(\tau/\bar\tau)^n = 1$; so the Weil classes are not in the N^n-eigenspace, and an N^n-eigenclass of degree $2n$ is a multiple of η^n (Fig. 8). The scaling of the secant sheaf by d^n under an isogeny of degree d (Lemma 1) is asserted without proof.

e. Step 4 of Theorem 4. The points of a Shimura variety with smaller Mumford–Tate group are the points of proper special subvarieties, most of which are not of CM type; and Pohlmann's theorem [34] covers the CM types whose Hodge ring is generated by divisors, not all CM abelian varieties.

f. The reduction to Weil classes (Theorem 3, Section 5). Moonen and Zarhin describe every Hodge class as a polynomial in divisor classes and Weil classes of isogeny factors in dimension at most five [32], and the description is not known beyond; simple CM abelian varieties can carry Weil classes [38, 45]; and not every factor of type IV is of Weil type.

g. Theorem 12. The Kuga–Satake construction exists for Hodge structures of K3 type, with $h^{2,0} = 1$ [12, 21]; Theorem 12 applies it to arbitrary primitive classes. Its algebraicity, credited there to Deligne, is open: Deligne proved that the correspondence is absolutely Hodge [13], and its algebraicity is known only in special families [40, 41]. Since the case split of Section 1.4 rests on this step for the classes of coniveau zero, it is incomplete.

None of these points is repaired here. The present paper does not use secant cycles, the eigencharacter argument or the Kuga–Satake construction outside K3 type, and it claims no case of the conjecture that is not proved in Sections III to VI.

VIII. WHAT IS PROVED AND WHAT IS OPEN

Table III lists the statements. Three remarks on what the table says.

The two open statements are not independent problems that happen to add up to the conjecture; they are two halves of one phenomenon. Both say that the algebraic locus of a flat Hodge class, a countable union of closed strata through a dense set of points of complex multiplication, is not meagre. For (F2) the dense set is supplied by explicit cycles at CM points and the first instance is a single class on an abelian fourteenfold or a single Ext group on an eightfold; for (F3′) the dense set is supplied by the K3 surfaces with complex multiplication and by the chains of Section V, and the first instance is a cubic real multiplication on one K3 surface of Picard number 13.

Each construction of this paper ends at a question with a definite answer. Whether some convolution of the pieces of Theorem VI.3 has $\dim \mathrm{Ext}^2 = 57$ is a finite computation in the Ext algebra of two sheaves on a fourfold. Whether the link ξ is algebraic is a question about one Hodge class on one abelian variety, equivalent by Theorem VI.9(v) to the exceptional classes together with one more cycle. Whether a correspondence acting on $T(S_\lambda)$ otherwise than by a similarity exists is a question about one K3 surface. A negative answer to the first closes that route, as the analogous question on E_0^6 closed; a positive answer to any of them proves the corresponding statement at its first instance.

A. Specialisation, degeneration and blow-ups

Three devices are often proposed for passing from the members where a class is known to be algebraic to the others, and it is worth recording exactly what each one proves.

Proposition VIII.1. *Let $\pi : \mathcal{X} \to B$ be a smooth projective family over an irreducible base and ξ a flat Hodge class on its fibres.*

(i) *If ξ_t is algebraic at the generic point of a curve $C \subseteq B$ then it is algebraic at every point of C. The converse implication, from one point of C to its generic point, holds when ξ_{t_0} is the Chern character of a perfect complex whose semiregularity map is injective, and in no greater generality is it known.*

(ii) *Let $\tilde X$ be the blow-up of a smooth projective variety X along a smooth closed subvariety Z of codimension c. Then $\mathrm{Hdg}^k(\tilde X)/\mathrm{Alg}^k(\tilde X) \cong \mathrm{Hdg}^k(X)/\mathrm{Alg}^k(X) \oplus \bigoplus_{i=1}^{c-1} \mathrm{Hdg}^{k-i}(Z)/\mathrm{Alg}^{k-i}(Z)$. In particular the Hodge conjecture for $\tilde X$ is equivalent to the Hodge conjecture for X and for Z, and no sequence of blow-ups and blow-downs reduces the defect of a variety.*

(iii) *For the Weil classes of $\mathcal{D}_{n,n}$ and the exceptional class of a Mumford fourfold, every degeneration to a fibre with a semistable model on which the limit class is algebraic has its lift to the generic fibre as its missing step, and for the Mumford fourfold no degeneration keeping the class of Hodge type exists, since the Hodge locus of the class is a compact Shimura curve.*

Proof. (i) Specialisation of cycles along a curve [14, Ch. 20] sends an algebraic cycle on the generic fibre to one on each special fibre with the same flat cohomology class; this is the direction used by Schoen's degeneration and by Proposition IV.6. In the other direction the algebraic locus is a countable union of closed sets by Lemma IV.2, and a point in one stratum says nothing about the stratum being the whole base; the one proved lift is the semiregularity theorem of Bloch, Buchweitz and Flenner [5, 6], which is the content of Theorem VI.1. (ii) Manin's decomposition of the motive of

TABLE III. Every statement of the paper with its status. "Proved" means proved in the text from the cited results; "open" means not proved here or elsewhere to our knowledge.

statement	where	status	what it needs
HC $\Leftrightarrow$ (F2) $\wedge$ (F3′)	Thm. I.1	proved	–
(F2) $\Leftrightarrow$ **AB**	Thm. III.7	proved	–
six minimal sufficient sets	Prop. III.8	proved, checked in Lean	–
dichotomy of the algebraic locus	Lem. IV.2	proved	–
(F2) as propagation from CM points	Thm. IV.5	proved	–
Weil loci dense with CM base point	Prop. IV.6	proved in [7]	–
chains make e with $e^2 \in \mathbb{Q}$ algebraic	Thm. V.3	proved	a chain of the right multiplier
$\sqrt{6}$ algebraic on two-dimensional families	Thm. V.5	proved	–
$\sqrt{5}$, $\sqrt{7}$ algebraic on one-dimensional families	Thm. V.8	proved	–
similarities act by multipliers $d_1 d_2$	Prop. V.9	proved	–
finite-group correspondences give abelian E	Prop. V.10	proved	–
HC for $X \times X$, X in the dihedral families	Cor. V.11	proved	–
(F3′) for S_λ, cubic E	Sect. V F	open	a non-similarity correspondence
(F3′) in general	–	open	–
product object of polarised shape	Thm. VI.3	proved	–
direct sums never semiregular	Prop. VI.4	proved	–
(P2′) at a product point	Sect. VI E	open	Ext* of secant sheaves
sixfold link $\Rightarrow$ exceptional classes	Thm. VI.9	proved	–
algebraicity of the link ξ	Sect. VI G	open	a cycle on $X \times X \times Z_K$
exceptional classes of $X \times X$	–	open	–
(F2) in general	–	open	–
the Hodge conjecture	–	open	(F2) and (F3′)

a blow-up [23], $h(\tilde{X}) = h(X) \oplus \bigoplus_{i=1}^{c-1} h(Z)(-i)$, is realised by the algebraic correspondences π^* and $j_* \circ p^* \circ$ (powers of the relative hyperplane class), which are compatible with Hodge classes and with algebraic classes; so both the Hodge classes and the algebraic classes of $\tilde{X}$ decompose accordingly, and the quotients add. A birational map between smooth projective varieties factors through blow-ups and blow-downs along smooth centres only in special cases, but every birational map is dominated by a sequence of blow-ups, and (ii) shows that the defect of the top variety is at least that of the bottom one, so the defect cannot be removed on the way. (iii) A flat class has a limit in the limit mixed Hodge structure of a semistable fibre, and on a totally degenerate fibre the graded pieces are combinatorial and every class is algebraic on the components; the return to the generic fibre is the lift of (i), with the same unproved step. For a Mumford fourfold the Hodge locus of the exceptional class is the compact Shimura curve of [7, Thm. 21.7], which has no boundary, so no fibre at which the class is still of Hodge type degenerates. □

The devices therefore relocate the two open statements without weakening them. The one true reduction in shape is Theorem IV.5: the whole of (F2) is equivalent to a lift from CM points along polarised abelian schemes, and the lift along each scheme is a question about one semiregular object at one point.

Nothing in the paper assumes an open statement, and no section depends on a computation that is not also proved by hand in the text. The machine checks of Appendix A recompute the finite quantities; they do not carry any proof.

Appendix A: Machine checks

Every finite quantity used in the proofs is recomputed by programs in the archive [4], and the combinatorial and arithmetic statements among them are also proved in Lean 4. No statement of the paper rests on these checks; they guard against slips in the hand computations.

a. Lean. The file `lean/HodgeClosure.lean` (Lean 4.24.0, no Mathlib) contains thirty-six theorems. Section 1 encodes the eight statements and the rule set of Table I as data, computes closures by iteration, and proves by evaluation that the closures of $\{\mathrm{F2}, \mathrm{F3}'\}$, $\{\mathrm{F3}\}$, $\{\mathrm{L}, \mathrm{M}\}$, $\{\mathrm{L}, \mathrm{F3}'\}$ and $\{\mathrm{V}, \mathrm{F3}'\}$ contain **HC**, that the closures of $\{\mathrm{F2}\}$, $\{\mathrm{F3}'\}$, $\{\mathrm{L}\}$, $\{\mathrm{M}\}$, $\{\mathrm{V}\}$, $\{\mathbf{AB}\}$, $\{\mathrm{L}, \mathrm{V}\}$ and $\{\mathrm{M}, \mathrm{F3}'\}$ do not, and that the list of minimal sufficient sets, computed from the rule set alone, has exactly six members, those of Proposition III.8. Section 2 works in $\mathbb{Z}[\sqrt{-d}]$ and proves that the hermitian form η transforms under τ by $N(\tau)$, that the norm is multiplicative, that η is not $K^\times$-invariant, and that the characters $N(\tau)^n$ and τ^{2n} differ at explicit τ (Section VII). Section 3 proves the dimension counts of Section VI: $3n < n^2$ for $n \geq 4$, $m^2 + m'^2 < (m + m')^2$ for $m, m' \geq 1$ with $2mm'$ mixed directions, the values $r = 57$ and 104 of the lower bound, and the Hodge class count 26 of Theorem VI.9(i) as a sum of Künneth contributions. Section 4 proves the polynomial identities behind the Fricke families of Section V D: $c_4^3 - c_6^2 = 1728\, t\, (t^2 + 13t + 49)^2$ at level 7 and

$1728\,t\,(t^2+22t+125)^3$ at level 5, by exact arithmetic on coefficient lists, together with the root products 7^2 and 5^3, the Euler numbers 12 and 24, the Shioda–Tate counts giving Picard number at least 16, the transcendental rank 6, and the dimensions $(n+1)/2$ of the dihedral Hecke algebras. Every theorem is checked by `#print axioms`: thirty-four use no axiom, and the two inequalities proved with the arithmetic tactic use `propext` and `Quot.sound`.

The Lean certificate of [7], `HodgeObstruction.lean` (Lean 4.34.0, four hundred and nine theorems, each axiom-free or depending on `propext` only), carries the finite computations of that paper; it is cited and not repeated here. The reduction of Section III uses only the hand-proved parts of [7].

b. Python. `code/sqrt6_chain.py` (sympy) implements Hilbert symbols and Hasse invariants exactly, verifies the product formula on random pairs, classifies $U_{\mathbb{Q}}$, $E_8(-1)_{\mathbb{Q}}$ and the invariant lattices, reproduces Varesco's embedding results for $\sqrt{2}$ and $\sqrt{3}$ as controls, searches the 165 sign-admissible small forms q_E over $\mathbb{Q}(\sqrt{6})$ (98 embed in V_3, 67 do not), exhibits the complements $\langle 1,-10\rangle$ and $\langle 1,-1,-1,-1,-3,-30\rangle$, and checks both orders of the chain: twenty-five checks. `code/fricke_families.py` computes the fibre types, Euler numbers and Picard bounds of the Fricke curves and their twists at levels 5 and 7, checks numerically that the roots of $j(t)=j(\tau)$ are carried by $t\mapsto N^k/t$ to the j-invariants of the N-isogenous curves, verifies the twist identity $u_{p,q}(w(t)) = pq\,u_{w(p),w(q)}(t)/t^2$, the Möbius transformations preserving the labelled points, the embedding of every rank-six form of signature $(2,4)$ in V_2 and V_3 after any scaling, and the Hecke algebras of $(D_n,\langle\tau\rangle)$ for $n=5,7,9,11$: forty-one checks. `code/product_object.py` builds the exterior-algebra model of $H^*(A\times A')$ for $(m,m')=(2,1),(2,2)$, the tangent space of $\mathcal{D}_{n,n}$ and of the product locus, the first-order Hodge loci of the exterior products and of the object of Theorem VI.3, the Vandermonde determinant, the vanishing of the cross terms, and the Hankel rank and $r(\gamma)$ of the final class; `code/convolution_checks.py` computes in the same model the rank of $v\mapsto v\lrcorner\theta_A$ on $T\mathcal{D}_{n,n}$, the contraction ranks of $\mathrm{ch}(\mathcal{E}')$ and $\mathrm{ch}(\mathcal{E}'')$ in Proposition VI.5(iii), and the trace of an upper triangular times a strictly upper triangular block matrix. In these two programs ranks are computed in floating point with a relative tolerance of 10^{-7} and agree with the closed formulas. `code/sixfold_link.py` checks the alternation and invariances of ψ_c, the decomposition of the K^1-invariants, the ratio of the induced forms at the three places, the symmetry of $T_X(-1)$, the count 26 and the weight support of ξ; `code/link_weights.py` computes the Hodge classes of degree four on $X\times X\times Z_K$ under the three groups of Section VI G and the vanishing of $\mathrm{Hom}(X_c,Z_{K,c})$ at the compositum CM point. `code/closure_graph.py` recomputes the closures and the minimal sets, and `code/bypass_checks.py` the Horn closure with the facts of the literature on abelian varieties adjoined and the blow-up bookkeeping of Proposition VIII.1(ii).

c. Julia, C, C++ and shell. `code/closure_graph.jl`, `code/closure_graph.c` and `code/closure_graph.cpp` recompute the closures of all 256 subsets of the eight statements and the six minimal sets by three independent implementations of the fixed-point iteration; `code/eigencharacter.jl` recomputes the characters of Section VII in exact integer arithmetic, and `code/hilbert.c` the Hilbert symbols and the invariants of Table II; `code/run_all.sh` runs every program and the Lean file and prints a line per check. The expected outputs are recorded in `code/EXPECTED.md`.

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DATA AND CODE AVAILABILITY

The LaTeX source of the paper, the Python, Julia, C, C++ and shell programs of Appendix A and the Lean 4 file are archived at Zenodo under `doi:10.5281/zenodo.XXXXXXX` [4]; the compiled paper is the only file of the folder `preprintHodgeConjecture` of that archive. Deep Bhattacharjee carried out the constructions on (F2) and (F3′) and the Lean, Julia, C, C++, Python and shell verifications with the help of AI assistance (Claude, Anthropic). Both authors approved the final version.

ETHICAL DECLARATIONS

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